

Algorithmic Underwriting in High Risk Mortgage Markets*

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Janet Gao
Georgetown University

Hanyi Livia Yi
Boston College

David Zhang
Rice University

Abstract

We study the effects of a policy that increased lenders' reliance on algorithmic underwriting for low-credit-score, high-leverage mortgages. The policy led to a substantial credit expansion with little change in default risk among approved borrowers, suggesting that algorithms perform effective credit screening. The effectiveness of algorithmic underwriting arises from alleviated regulatory concerns by lenders, reduced information frictions in financial intermediation, and a greater reliance on standardized data. The policy also enabled low-credit-score borrowers to move to higher-quality neighborhoods. However, the credit expansion was weaker for Black and lower-income borrowers, which highlights algorithms' limitation in promoting financial inclusion for disadvantaged populations.

Keywords: Algorithmic Underwriting, FinTech, Household Leverage, Racial Inequality in Mortgage Markets, Mobility, Financial Inclusion.

JEL classification: G18, G21, G51, O33

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1 Introduction

Policies targeting household lending must carefully balance the benefits of expanding financial inclusion for high-risk borrowers with the costs associated with increased default (Layton, 2023). Better access to mortgage markets for these borrowers can help reduce disparities in home ownership rates (Eggers, 2001). However, extending loans to high-risk borrowers could also elevate the risk exposure of financial institutions and government agencies. A crucial component that influences this trade-off is loan underwriting, where lenders collect and verify applicants’ documents, assess their credit risks, and decide on loan approval. While a task traditionally performed by humans, underwriting has become increasingly automated over the past decades. By the mid-2000s, nearly all lenders were using automated underwriting systems (AUS) in some aspects of their lending practices (Wells, 2023).

How does the increasing reliance on algorithmic underwriting affect the trade-off between financial inclusion and risk management? The answers to this question are not obvious *a priori*. On the one hand, algorithmic underwriting faces limitations in collecting and interpreting soft information, which may affect its ability to evaluate the credit risks of borrowers with unconventional income and opaque credit history. On the other hand, compared to humans, algorithms are potentially less prone to errors and less influenced by agency conflicts or regulatory risk concerns. Despite the prevalence of AUS, limited empirical evidence exists regarding its role in high-risk market segments.

This paper studies the consequences of increased reliance on algorithmic underwriting in a low-credit-score, high-leverage segment of the US mortgage market. We examine the effects of a policy change implemented by the Federal Housing Administration (FHA) in August 2016, which targets borrowers with credit scores below 620 and debt-to-income (DTI) ratios above 43%. Before this date, the FHA mandated manual underwriting for all such borrowers; after August 2016, the FHA allowed AUS to approve loans directly without human underwriting. Understanding the role of algorithmic underwriting in this context is economically important, as the FHA is the primary source of mortgage insurance for low-credit-score and high-leverage borrowers and insures a \$1.5 trillion portfolio. We study the policy’s impact on loan quantities, performance, prices, and household mobility. We find that the policy leads to a substantial expansion of credit for affected borrowers with little change in their delinquency rates. At the same time, such expansion is limited within Black and lower-income borrowers. The credit expansion in turn allows low-credit-score individuals to relocate to higher-quality neighborhoods. Our findings highlight the ability of algorithmic underwriting to enhance financial inclusion and effectively manage credit risk, as well as its limitations in promoting financial access to disadvantaged populations.

We assemble a large dataset to address our research questions, starting with individual loan-level data provided by the Government National Mortgage Association (“Ginnie Mae”). This database covers the

near-universe of FHA-insured loans, and includes information on loan contract terms such as interest rates, amount, maturity, and purpose; borrower and property information such as the locations of purchased properties, borrower credit scores, and debt-to-income ratios; and importantly, information on loan delinquency. We merge this data with the Home Mortgage Disclosure Act (HMDA) data using the FHA endorsements as the intermediate link. This merge allows us to observe borrower income and race/ethnicity. We track the residential locations of individuals from a 1% randomized sample from Experian to measure household mobility. Finally, we use intergenerational mobility measures from [Chetty et al. \(2025\)](#) to measure neighborhood quality.

We begin by analyzing changes to the quantity of loans around the adoption of the policy. We focus on loans with DTI ratios up to 50 to compare human and algorithmic underwriting decisions, since human underwriters were not allowed to underwrite loans with above-50 DTI ratios pre-policy. Initial analysis suggests a substantial increase in the number of loans issued to affected borrowers, who have credit scores below 620 and DTI ratios between 43 and 50 (i.e., “high-DTI” loans). We employ a counterfactual estimation approach to draw causal inferences regarding the effects of the regulation change ([DeFusco et al., 2020](#)). This approach utilizes high-credit-score borrowers, who are unaffected by the policy, as the control groups, and uses the changes in the DTI distribution among this group as the counterfactual for the changes among the affected group. Using this approach, we find that the policy reform increases the total quantity of high-DTI loans taken out by low-credit-score borrowers by 6.5%, with a 122% increase in the high DTI region. We validate that the control group generates a reasonable counterfactual distribution by using multiple placebo analyses set in years with no policy change.

Given the large increase in credit quantity, a question naturally arises as to whether the increased reliance on algorithmic underwriting increases borrowers’ default probabilities. To answer this question, we compare the changes in delinquency rates following the policy event between treated (low-credit-score) and control (high-credit-score) borrowers, in both a difference-in-differences and triple-difference framework. We do not find evidence that delinquency rates increase more for treated borrowers. The magnitudes of the coefficients are small compared to the average delinquency rate of 16.3% for low-credit-score, high-DTI loans pre-policy. A remaining concern is that AUS may grant credit to “fragile” borrowers, who are prone to defaults when economic conditions worsen. To address this concern, we show that delinquency rates do not increase even in areas with the largest increase in unemployment rate during the sample period.¹ The results are also robust to alternative measures of delinquency rates, including less severe delinquencies and delinquencies over longer horizons covering the COVID-19 forbearance period. Combined, these results suggest that an increased reliance on algorithmic underwriting need not be associated with an increase in default risk among

¹Our analysis on unemployment rates is motivated by the recent evidence that the majority of mortgage defaults are driven solely by negative life events ([Ganong and Noel, 2023](#)), and the fact that defaults peak in areas with worsening economic conditions.

the approved borrowers.

We next explore how the policy-induced credit expansion varies across borrower race and income, motivated by the ongoing policy-relevant discussion regarding the unequal access to credit across race and income groups, and the potential role of algorithmic underwriting in moderating such inequality (Fuster et al., 2022; Das et al., 2023). Despite the policy’s focus on low-credit-score borrowers and FHA’s prevalence among minority borrowers, we find that the credit expansion is weaker among Black borrowers compared to non-Black ones, and among lower-income borrowers compared to higher-income ones. These differences suggest that the AUS may face limitations in expanding credit supply for historically disadvantaged populations.

We then investigate the mechanisms through which the FHA policy generates the large credit expansion by examining the heterogeneous changes in credit supply across various lender and borrower characteristics. Our results indicate that algorithms expand credit by easing lenders’ regulatory concerns, reducing information asymmetries along financial intermediation chains, and increasing the reliance on standardized data.

We start by assessing the regulatory concerns channel, i.e., lenders are concerned about the FHA denying insurance claims and closely monitoring their underwriting practices, as well as the risk of investigations by other government agencies over potential fraudulent activities. For example, the Department of Justice litigated several large lenders over their FHA insurance practices over the 2010s (Frame et al., Forthcoming). Even without litigation, the FHA can deny insurance claims if it finds deficiencies in lenders’ underwriting process. Such concerns may have led human underwriters to implement overly conservative lending standards, but should be less relevant when loans can be directly approved by FHA-endorsed algorithms. To evaluate this explanation, we compare the effects of the policy adoption across various lender types. First, we compare banks and nonbanks, expecting nonbank lenders to have greater risk tolerance and lower regulatory concerns (Benson et al., 2023b). Second, we compare banks that have or have not faced litigation regarding FHA insurance practices. Consistent with the prediction under the regulatory concerns channel, we find that the FHA policy leads to significantly larger credit expansions for bank (13%) than nonbank (5%) lenders. Within bank lenders, we find a greater credit expansion from those with litigation records.

Next, we discuss the role of algorithms in affecting information frictions in the financial intermediation chains. In this discussion, we focus on the role of aggregators and their concerns regarding regulatory risk and elevated servicing costs due to deficient underwriting. Aggregators purchase loans from lenders (i.e., indirect sellers) and in turn securitize the loans. In 2022–2024, indirect sellers originated 32% of loans in the FHA mortgage market. While aggregators do not underwrite loans themselves, they are legally liable for delinquencies of the loans they originate and often need to bear the elevated servicing costs (Benson et al., 2023a). This separation of legal responsibility and control, compounded by the fact that aggregators do not perfectly observe the underwriting quality of indirect sellers, could add to the intermediation costs of aggregators. A greater reliance on FHA-endorsed algorithms should reduce aggregators’ concerns associated

with unobservable human underwriting quality. Consistent with this prediction, we find a significantly stronger credit expansion from indirect sellers, i.e., lenders that sell to aggregators, compared to direct originators. These results suggest that algorithmic underwriting can reduce information frictions along the financial intermediation chain and facilitate liquidity-provision along the chain.

Furthermore, we explore the role of hard information and standardized data in explaining our effects. Specifically, AUS may have an advantage in processing standardized, structured borrower information or rely more on such information in its decision-making than do human underwriters. This mechanism suggests that the FHA underwriting policy should expand credit supply to a greater extent for borrowers with more standardized, structured information, such as payment histories and verified income. Consistent with this prediction, we find a stronger credit expansion for repeat home buyers than for first-time buyers, as repeat buyers have a history of mortgage payments. We also find a greater credit expansion for borrowers with co-applicants than for sole applicants, given that co-applicants' credit and payment history serve as additional data points. In contrast, we document a smaller credit growth in areas with higher self-employment rates, where income is often irregular and may require more human information collection or discretion.

We also discuss and rule out some alternative mechanisms, such as a capacity constraints channel and a lender entry channel. The capacity constraints channel states that our results may be driven by algorithms' ability to expand lenders' capacity to process applications. Prior to the policy, human underwriters may have faced capacity constraints in processing large quantities of data. Allowing algorithms to directly approve loans alleviates the workload of human underwriters. Under this channel, lenders facing more severe capacity constraints should expand credit quantity more compared to lenders facing less constraints. We do not find more constrained lenders to increase credit supply more, suggesting that our results are unlikely to be explained by algorithms alleviating lenders' capacity constraints. We further show that lender entry only accounts for a limited fraction of credit expansion, thus is unlikely to drive our results.

Taken together, our evidence suggests that algorithms can reduce regulatory concerns and the opacity of underwriting quality along financial intermediation chains, and may have advantages in processing standardized data. These results highlight the strengths and limitations of the FHA's underwriting algorithm and can inform the future design of algorithms in credit underwriting. Moreover, to the extent that regulatory concerns, securitization, and data availability concerns are common features of modern credit markets, our findings are generalizable beyond the FHA setting and can potentially shed light on how regulator-endorsed algorithms shape equilibrium outcomes in other markets.

What are the consequences of the FHA underwriting policy on households' borrowing costs and mobility? We start by investigating the financial consequences, i.e., whether high-leverage borrowers experience a change in borrowing costs as a result of the policy change. We find no change in interest rates for high-DTI loans, and an economically small and statistically weak increase in interest rates for low-DTI loans. One

potential explanation for this finding is the changes in borrower composition: as higher-income borrowers increase leverage and move to the high-DTI category, lenders may consider the remaining low-DTI borrowers to be riskier than before, thus charging higher rates.²

We next examine the non-financial consequences of algorithmic underwriting for households, focusing on whether policy-induced credit expansion facilitates moves to neighborhoods associated with greater economic opportunity. We use measures of intergenerational upward mobility from [Chetty et al. \(2025\)](#), which capture the expected income rank of children from low-income families across neighborhoods. We implement a two-stage least squares (2SLS) framework. In the first stage, we regress an indicator for obtaining a new FHA mortgage on an individual’s treatment status, defined as having a credit score below 620 interacted with a post-policy indicator. In the second stage, we relate changes in neighborhood quality to the predicted probability of FHA mortgage uptake. Our estimates suggest that low-credit-score individuals are more likely to obtain an FHA mortgage after the policy change, and this increased mortgage access is associated with moves to neighborhoods with higher upward mobility. To further address concerns that our results might be explained by unobserved differences between low- and high-credit-score individuals, we design a placebo analysis using individuals who remain renters throughout our sample period. We find no economically meaningful change in neighborhood quality for these individuals, suggesting that the take-up of FHA mortgages is a key mechanism for the improvement of neighborhood quality.

Taken together, the evidence is consistent with mortgage access playing an important role in households’ ability to “move to opportunity.” Two limitations are worth noting. First, because we cannot observe DTI in the credit bureau panel until an individual takes out a mortgage, we cannot use the earlier bunching approach, which requires an additional layer of inference based on high- and low-DTI bin comparisons and could be considered more rigorous. Second, these migration estimates capture local average treatment effects for high-leverage, high-risk households and may not generalize to the broader population or other policy contexts.

Finally, we quantify the potential effects of the policy on FHA insurance premiums. We note that while we do not detect an increase in the average delinquency rates within the affected group of low-FICO, high-DTI borrowers, the policy leads to an influx of such borrowers, which could increase the overall risk of FHA-insured loans. Through a back-of-the-envelope calculation, we infer that the policy increases the delinquency rate per dollar of FHA loans by 1.28%. To break even on the increase in credit risk, the FHA needs to raise the annual mortgage insurance premium (MIP) by 1.41 bps for an average 7-year mortgage. This means that, while the policy improves the financial inclusion for higher-risk borrowers, it could impose some costs on lower-risk ones.

Our study contributes to several strands of literature. First, we add to the burgeoning literature analyzing

²While the reduced reliance on manual underwriting may have reduced labor cost for lenders, such cost-saving is likely small due to the rigidity in the labor markets. Lenders’ market power may also limit the pass-through of lower costs to loan pricing. Both of these reasons could explain why we do not detect an increase among high-DTI borrowers.

the increasing use of technology in mortgage underwriting (Foote et al., 2019; Fuster et al., 2019, 2022; Johnson, 2023b; Das et al., 2023). Existing evidence suggests that human loan officers are influenced by volume-based incentives and errors (Tzioumis and Gee, 2013; Cortés et al., 2016; Giacoletti et al., 2023), while human risk managers can reduce delinquency (Berg, 2015). In comparison, Fuster et al. (2019) find that FinTech lenders process loan applications faster and respond more elastically to demand shocks. Other literature shows that increased lender automation can engender differential impact on the credit access across racial and gender groups, though the direction of the change depends on the context (Dobbie et al., 2021; Fuster et al., 2022; Bartlett et al., 2022; Erel and Liebersohn, 2022; Chu et al., 2023; Das et al., 2023; Howell et al., 2024). We complement this literature by examining the effect of algorithmic underwriting when human judgment is still present and used as a complement. Our results suggest that under some extent of human supervision, algorithmic underwriting can simultaneously allow for a large increase in credit supply with little change in loan default probabilities in the low-credit-score, high-leverage segment of mortgage markets. However, consistent with earlier evidence, we find that not all borrower groups benefit equally from this credit expansion.

Second, our paper is related to the broader literature on algorithmic underwriting in financial intermediation. In particular, in the auto loans market, Jansen et al. (2024) use a randomized experiment to show that algorithmic underwriting outperforms human underwriting for riskier and more complex auto loans.³ Costello et al. (2020) study the implications of AI-based lending models for trade creditors. In a developing country context, Tantri (2021) and D’Acunto et al. (2022) study the role of algorithms in increasing efficiency and reducing discrimination in personal loans and peer-to-peer loans, respectively. We study the role of human-augmented algorithmic underwriting in the U.S. mortgage market, which has been particularly controversial due to its size and importance (Fuster et al., 2022; Das et al., 2023). Our findings in this market can inform a broader set of consequences of algorithm-based lending for households and government agencies, such as the capacity for financial inclusion of high-risk borrowers and their subsequent location choices, the risks borne by government agencies, and the distributional consequences of the credit expansion across income and racial groups.

Finally, our paper complements and expands the existing literature on the effects of household leverage policies. DeFusco et al. (2020) show that the Dodd-Frank “Ability-to-Repay” rule, which imposes restrictions on high DTI lending, led to a reduction in credit supply but had limited effects on mitigating default risks. Following their methodology, we analyze the bunching of mortgage loans around regulatory thresholds. Other studies based in the U.S. suggest that DTI restrictions not only directly affect house prices, but also generate spillover effects on groups that fall outside the established limits (Foote et al., 2010; Greenwald, 2018; Johnson, 2020, 2023a). Beyond the U.S. context, several studies further examine the implications of

³Jansen et al. (2024) find that algorithms approve fewer auto loans, but charge higher interest rates and are associated with lower default rates. In contrast, we find algorithm underwriting is associated with significant credit expansion but little change in interest rates or delinquency in the mortgage market.

household leverage regulations for housing choices, household leverage, mortgage credit supply, and house prices (Van Bekkum et al., Forthcoming; Kinghan et al., 2022; Acharya et al., 2022; Tzur-Ilan, 2023; Laufer and Tzur-Ilan, 2021). Unlike the policies studied in prior work, the policy we analyze emerges from the variation in the relative weight of algorithms and human involvement in the underwriting process.

2 Institutional Background

To qualify for FHA insurance, mortgage lenders must abide by the FHA underwriting guidelines. The guidelines stipulate that all transactions, with certain exemptions, must be scored through the Technology Open To Approved Lenders (TOTAL) Mortgage Scorecard (see FHA Single Housing Policy Handbook 4000.1, Section II (A) (4)). The TOTAL Mortgage Scorecard is an algorithm introduced by the U.S. Department of Housing and Urban Development (HUD) in 2000 to assess the creditworthiness of mortgage applicants and predict mortgage default. The Scorecard accounts for over a hundred data elements as input, including the applicant’s monthly income, house appraised value and sale price, loan amount, loan-to-value ratio, front-end and back-end DTI ratios, and more.⁴ The Scorecard is designed to streamline the underwriting process and provide lenders with a quick and consistent evaluation of borrowers’ creditworthiness.

The TOTAL Scorecard provides two process classifications: “Accept” or “Refer.” Accept implies that the system determines that the borrower meets the FHA’s underwriting guidelines and is eligible for an FHA-insured loan. This means the borrower’s application can move forward in the approval process. Refer means that the information provided by the borrower is not sufficient for the system to make a clear decision. This occurs when the automated underwriting system finds the borrower eligible but cannot determine an approval. All mortgage applications that receive a “Refer” classification are required to go through manual underwriting before a final decision on the application is rendered (FHA’s Office of Single Family Housing Training Module 4).

The FHA’s Training Module specifies that the human underwriting process mainly consists of evaluating the following categories of information: (1) Borrowers’ credit history: The human underwriter should check whether borrowers exhibited multiple recent late payments (e.g., more than two 30-day delinquencies for housing and installment loans over the past 12 months and more than two 60-day delinquencies for revolving loans over the past 24 months), and if such late payments exist, whether they reflect extenuating circumstances or “a disregard for financial obligations” or “an inability to manage debt.” (2) Housing obligation payment history: The human underwriter must verify and document receipt of housing obligation payments over the past 12 months from various sources, including credit reports, receipts of rent from landlords, or

⁴A complete list of data elements can be found in Appendix A of the AUS Developer’s Guide: <https://apps.hud.gov/pub/chums/aus-developers-guide-SOAP-MISMO.pdf> (accessed April 2024).

documentation that the applicant lived rent-free over the time period. (3) Liabilities and debts: The human underwriter must quantify the liabilities in the credit report plus any additional monthly obligations revealed during the application process (such as alimony payments and federal debts), and verify that they satisfy the FHA's qualifying ratios. (4) Income: The human underwriter must collect information on whether the borrower had frequent changes in employment, gaps in employment, and consider factors such as maternity leaves and expected raises that will be received within 60 days of mortgage closing. The DTI ratio for human underwritten loans must not exceed 50.

Overall, the human underwriting process involves information acquisition, verification against hard criteria, and human judgment. The FHA training module particularly emphasizes that human underwriters need to exercise discretion in the process, by noting that “[a] mortgage transaction that meets minimal standards does not necessarily constitute prudent underwriting. The underwriter should judge the overall merits of the loan application to render a justifiable and documentable decision.”

Following the financial crisis, regulators have increased their focus on risk management in the U.S. mortgage market, including the creation of Dodd-Frank Act provisions targeting household leverage (DeFusco et al., 2020). Consistent with this trend, effective April 2013, HUD updated the TOTAL Mortgage Scorecard to include a manual underwriting mandate for FHA borrowers with credit scores below 620 and debt-to-income ratios exceeding 43.00% (Mortgagee Letter 2013-05). This change meant that borrowers falling into this category could not receive an “Accept” recommendation from the TOTAL Scorecard but would be downgraded to a “Refer” recommendation, and have to go through human underwriting to receive FHA coverage. However, this policy had little immediate practical effect because FHA loans with credit scores below 620 were already rare following the financial crisis, likely due to the FHA's rules in evaluating lenders.⁵ In August 2015, the FHA implemented a Supplemental Performance Metric that made it more feasible, in principle, for lenders to originate loans to low-credit-score borrowers.⁶

The manual underwriting mandate was lifted in August 2016 for FHA borrowers with credit scores below 620 and DTI ratios above 43%.⁷ While the reason for the change was not announced in a Mortgagee Letter, it was likely due to considerations of financial inclusion. Under the revision, borrowers in this category could once again receive “Accept” recommendations from the TOTAL Scorecard if they were determined to be creditworthy by the automated underwriting system. The TOTAL Scorecard Version 3 underwriting algorithm, which is machine-learning based, was applied throughout our study period without major changes.⁸

⁵See a description of the problem facing low credit score borrowers [HERE](#) and FHA's request for comments [HERE](#).

⁶See the policy fact sheet [HERE](#).

⁷See the description of the policy change [HERE](#) and the corresponding FHA TOTAL Scorecard update on page E-17 [HERE](#). As described in the first reference, in March 2019, the FHA partially reinstated this policy by referring more high-risk borrowers, especially those with multiple risk factors, to manual underwriting, but the volume impact of this partial reinstatement to credit score under 620, DTI over 43 borrowers was small as can be seen in Figure 1.

⁸See a description of the TOTAL scorecard and its changes [HERE](#). The updates to its algorithm occurred in 2008, when version

We study the effects of the expanded use of algorithmic underwriting in August 2016 on credit supply and default risk. This policy change only affected highly levered, low-credit-score borrowers. Borrowers whose credit scores are above 620 and DTI significantly below 43 were not affected and can serve as the “control groups” in our analysis.

There are limited alternative mortgage options available to our treated group of low-credit-score borrowers during our sample period. GSE mortgages are not available to this group of borrowers. Subprime private label securitization was common before the financial crisis but its volume fell sharply in 2007-2008 ([Frame et al., 2021](#)). While in theory portfolio lending is a possible alternative to FHA lending, [Kim et al. \(2024b\)](#) show that such lending is minimal for low-credit-score or highly levered borrowers. This means that any changes in FHA credit that we measure likely capture the overall changes in mortgage credit available to low-credit-score borrowers.

Throughout our study period, lenders have an incentive to screen borrowers against their default risk, for at least two reasons. First, in the event of an FHA borrower delinquency, the cost of loan servicing can rise significantly.⁹ Second, if lenders submit claims to the FHA for reimbursement for defaults, they face the risk of the FHA discovering underwriting mistakes on the defaulted loans and holding them liable for the damages (see, e.g., [Parrott and Goodman, 2019](#); [Frame et al., Forthcoming](#)). This incentive of lenders to screen borrowers, despite their principal loss being insured, has also been found to be important in the GSE and private label securitization markets ([Bosshardt et al., Forthcoming](#); [Adelino et al., 2025](#)).

3 Data and Variables

3.1 The Ginnie Mae-HMDA Matched Sample

Our analysis primarily relies on a Ginnie Mae-HMDA matched sample. Ginnie Mae guarantees timely principal and interest payments for FHA-insured mortgages and publicly discloses the origination and performance of loans included in its MBS issues starting in September 2013. The Congressional Budget Office (CBO) estimates that the Ginnie Mae MBS issues make up about 97% of FHA-insured mortgages.¹⁰ The loan-level disclosure data we use is comparable to the data compiled by eMBS, which has been used in

2 was introduced, and in 2012, when version 3 was introduced. Anecdotal evidence suggests that the machine-learning algorithm in question may be a logistic regression ([Fuster et al., 2022](#)).

⁹As explained in [Goodman \(2014\)](#): “The costs of servicing delinquent loans are much higher than the costs of servicing performing loans. [...] According to MBA estimates, non-reimbursable costs and direct expenses associated with the FHA’s foreclosure and conveyance policies were two to five times higher than for GSE loans, even before the GSEs changed their compensatory fee schedule. In 2013, the annual cost of servicing a nonperforming loan was on average 15 times that of servicing a performing loan—\$2,357 versus \$156.”

¹⁰See the breakdown [HERE](#).

a number of recent studies including [Fuster et al. \(Forthcoming\)](#) and [Kim et al. \(2024a\)](#), with the latter describing it as “essentially [...] the entire universe of FHA and VA mortgages.”

The Ginnie Mae loan-level database contains a rich set of underwriting information including the debt-to-income ratio, credit score, property type, and loan purpose, and loan characteristics including the interest rate on the mortgages, the upfront and annual mortgage insurance premium (MIP), the loan amount, loan term, whether the mortgage is fixed-rate or an ARM, and the month of origination are also observed in the data. Furthermore, it contains information about the delinquency status of the mortgages.

We focus on new purchase mortgages and exclude streamline refinances, which have missing debt-to-income ratios and credit scores in the Ginnie Mae data. We further restrict the sample to fixed-rate, single-family, non-manufactured housing mortgages, which is the predominant form of FHA-insured mortgage lending during our sample period.

A limitation of the Ginnie Mae data is that it does not include information about the income and demographics of the borrower, the borrower’s geographical location beyond state, or the borrower’s race and ethnicity. We obtain these variables from the 2013–2017 Home Mortgage Disclosure Act (HMDA) data, and merge such information with the Ginnie Mae data via the publicly available FHA Single-Family endorsements data. Our matching process relies on variables such as the interest rate on the mortgage, the month of the endorsement, and the property zip code. Details of this data and the matching procedure are provided in Internet Appendix [A](#), together with results showing the robustness of our base findings using the Ginnie Mae sample before being merged with HMDA.

The merged Ginnie Mae-HMDA database allows us to examine the change in origination volume in months around the FHA policy change. Our analysis focuses on the two-year window centered around August 2016, i.e., August 2015 to August 2017, excluding the month of the policy change (August 2016). Our analysis focuses on loans with DTI ratios below 50, given that human underwriters were not allowed to approve loans with above-50 DTI ratios. In Internet Appendix [B](#), we discuss and explore the above-50 DTI segment of the market.

To compute interest rate spreads, we take the difference between the mortgage interest rate and the Freddie Mac Primary Mortgage Market Survey Rate (PMMS) during the month of origination.¹¹ Delinquency rates refer to the 90-day delinquency within two years of origination in our baseline analysis, which we expand by time horizon in robustness checks. We also control for loan characteristics such as the log of loan amount and the log of borrower household income in some specifications.

¹¹ Available at: <https://fred.stlouisfed.org/series/MORTGAGE30US>.

3.2 Experian Data

We track households' changes in address using data from Experian, a major credit bureau in the U.S. The data contain a 1% national sample of U.S. individuals who are randomly selected based on the last two digits of their social security number. The dataset describes detailed individual demographic and economic characteristics, such as the individual's address (accurate to the census tract), age, sex, marital status, credit score, estimated income, and debt characteristics by category (auto, mortgage, credit card, student loan, medical debt, etc.).

We build an individual-year panel using annual Experian data from 2013 to 2019. We exclude the year 2016 from our sample because it includes both pre- and post-treatment periods. This panel dataset allows us to track individual addresses (accurate to the census tract) over time, thus identifying movers in a given year. The detailed debt variables also allow us to identify those who obtain a new FHA mortgage in a given year. To measure *changes* in neighborhood quality, the regression sample is limited to years 2014-2019. We include controls for individual characteristics such as gender, marital status, and credit score.

We further enrich the Experian individual-year observations that include new FHA mortgage originations by merging with the Ginnie Mae data via the publicly available FHA Single-Family endorsements data. The details of the enrichment merge and the match rates are described in Internet Appendix A.3.

3.3 Neighborhood Quality

We assess neighborhood quality based on intergenerational upward mobility. Upward mobility metrics capture the household income rank for children of parents at the 25th and 50th percentiles. They are measured at the census-tract level and come from Chetty et al. (2025). In robustness analyses, we look at additional census-tract level measures of neighborhood quality that have strong correlations with upward mobility, including public school ratings, average household income, poverty rate, share of college-educated residents, and share of single-parent households. Public school ratings are obtained from GreatSchools.org, and other neighborhood quality attributes come from the Opportunity Insights website.¹²

All neighborhood quality measures are merged with the Experian data using individuals' census tract or zip code. If a zip code spans multiple school districts, we assign it to the district covering the largest share of its population based on Census crosswalks. Using the merged dataset, we compute the year-on-year change in an individual's neighborhood quality.

¹²https://opportunityinsights.org/data/?geographic_level=0&topic=0&paper_id=1652#resource-listing (accessed July 2025).

3.4 Other Variables

We examine the heterogeneity of effects across various characteristics of the borrower, lender, and local markets. From the borrower’s side, we observe multiple characteristics including race, income, whether a borrower is a first-time or repeat home buyer, and whether a borrower is the only applicant or has co-applicants. For race, we separate our sample into Black and non-Black borrowers. For income, we partition the sample into high- and low-income groups according to whether borrowers’ relative household income exceeds the sample median. Relative household income is defined as the ratio of household income over the MSA median. This adjustment helps us compare across borrowers within the same broad geographical area, instead of comparing across those in far-apart regions, such as the Northeast vs. the Southwest. In addition, we consider whether a borrower is likely to be self-employed. Given that we do not have direct indicators for individuals’ self-employment status, we approximate its likelihood based on the percentage of income in a zip code arising from self-employment income. The fraction of self-employment income comes from the IRS SOI data.

From the lender’s side, we construct measures of regulatory concerns, lender types indicating their position in the financial intermediation chains, and capacity constraints. To characterize lenders’ regulatory concerns, we separate bank and non-bank lenders, where banks are defined as HMDA lenders that can be linked to an entity in the commercial bank Call Report through Robert Avery’s crosswalk.¹³ Other lenders are classified as nonbanks. FinTech lenders, as defined in [Fuster et al. \(Forthcoming\)](#), are not present in our market before or after the policy change. We also obtain from [Frame et al. \(Forthcoming\)](#) a list of lenders that faced litigation for alleged FHA insurance fraud from the Department of Justice. Next, we identify lenders that rely on aggregators (i.e., “indirect sellers”) as those who sell more than 95% of their originated FHA mortgages, excluding those held in their portfolios, to entities other than Ginnie Mae or affiliated institutions at the time of HMDA reporting. These lenders likely never sell to Ginnie Mae directly but rather rely on intermediaries who sell to Ginnie Mae on their behalf. Finally, we measure capacity constraints in two ways. First, we compute the year-on-year growth in application volume in each lender-zip-year, and partition the sample according to whether a market’s application growth is above the overall sample median. Second, we follow [Jo and Liu \(2024\)](#) to define six months with above (below)-median seasonal factor as the high (low) home purchase season.

3.5 Summary Statistics

Table 1 presents the definitions and summary statistics of the variables used in our study. The average loan in our sample has a 6-percentage-point delinquency likelihood and an interest rate spread of 14 basis

¹³ Available for download [HERE](#).

points, measured as the difference between the mortgage interest rate and the 30-year Freddie Mac survey rate. A typical borrower has a household annual income of \$71,646, and 12% of borrowers are Black. In the individual-year panel derived from the Credit Bureau data, the average likelihood of obtaining an FHA mortgage for a new home purchase is around 0.1% in a given year, and the average upward mobility ranges between 44-51%.

TABLE 1 ABOUT HERE

4 Effects on the Quantity of Credit

We first focus on how the FHA policy change affects the quantity of home purchase loans granted to households. We start by providing descriptive evidence on the changes in mortgage volume and then perform bunching estimation to sharpen causal inferences and quantify the shift in household leverage.

4.1 Descriptive Evidence

We first visually inspect how the quantity and composition of mortgage credit change around the FHA policy reform. We plot the percentage of mortgage loans whose DTI ratios exceed 43% (i.e., high-DTI loan share) for borrowers below and above the 620 credit score cutoff, respectively. Figure 1 depicts these statistics. The red, dashed (blue, solid) line represents the percentage of mortgages issued to high-DTI borrowers among the ones with below-620 (above 620) credit scores. The vertical line indicates the month of the FHA removal of human underwriting requirement, i.e., August 2016. The two lines evolve in parallel prior to the policy reform, exhibiting little pre-event trend. In the pre-policy period, high-DTI loans account for around 8% of the total number of mortgage loans extended to low-credit-score borrowers. This fraction jumps sharply after the policy date, rising to 18.6% within two months and nearly 29% after six months. In contrast, there is no abrupt change in the high-DTI loan share among high-credit-score borrowers (i.e., FICO>620, the control group).

FIGURE 1 ABOUT HERE

We next look into how credit growth following the policy reform varies around the 43% DTI cutoff. To do so, we compute the loan growth rate (i.e., change in log number) from the 12-month pre-event window to the 12-month post-event window. This growth rate is computed separately for each DTI integer category (i.e., 20, 21, 22, ..., 49, 50) for low- and high-credit-score borrowers, respectively. Figure 2 reports the results. The horizontal axis represents DTI ratio in integer percentage points. We find that for low-credit-score borrowers, loan growth rates hover around zero for DTI ratios below 35, and become negative for DTI between 36 and 43.

Above the 43 threshold, loan growth turns positive and economically large. The graphical evidence yields several implications. First, the policy change has little impact on low-leverage borrowers, whose DTI lies below 35. Second, it seems to reduce the number of borrowers taking out mortgages right below the 43 DTI threshold, and most importantly, increase the number of borrowers whose leverage exceeds the threshold. The tremendous growth of the high-leverage loans likely consists of both the switching of borrowers from below to above the 43 DTI threshold, and the influx of new, high-leverage borrowers in the market.

FIGURE 2 ABOUT HERE

In Figure B.2 of the Internet Appendix, we present the number of loans for each DTI grid extended to high- and low-credit-score borrowers around the policy change. Similar to Figure 2, there is a significant increase in low-credit-score, high-DTI loans after the policy. In contrast, there is little change in the DTI distribution among high-credit-score borrowers.

4.2 Bunching Estimator

To sharpen our causal inferences, we adopt the empirical design developed in DeFusco et al. (2020) to quantify the changes in FHA credit. The core idea behind this design is to construct a counterfactual DTI distribution for low-credit-score borrowers in the absence of the policy change, and compare the actual DTI distribution with this counterfactual. In our setting, high-credit-score borrowers are not affected by the policy change, so the changes in DTI distribution among these borrowers are considered as the counterfactual case. At each DTI level, we compute the counterfactual fraction of loans among low-credit-score borrowers by summing up two parts: (1) the pre-policy fraction of loans among low-credit-score borrowers, and (2) the changes in the fraction of loans among high-credit-score borrowers (i.e., counterfactual growth).¹⁴

Notations and Assumptions

Before describing our methodology, it is useful to introduce some notations. We use n_d to represent the actual number of loans within DTI integer bin d . Subscripts h and l indicate borrowers with credit scores above or below 620. Superscripts *pre* and *post* indicate event periods before and after the policy change.

Thus, n_{hd}^{pre} and n_{hd}^{post} represent the actual number of loans among high-credit-score borrowers for DTI integer bin d before and after the policy event, respectively. Similarly, n_{ld}^{pre} and n_{ld}^{post} represent the actual number of loans among low-credit-score borrowers at DTI bin d before and after the policy event. $\hat{n}_{ld}^{post}$ denotes the *counterfactual* number of loans among low-credit-score borrowers for DTI bin d after the policy event.

¹⁴This approach is modified from the standard bunching approach developed in the public finance literature, which involves fitting a polynomial to the observed distribution of a “running variable” while omitting the data immediately above and below the threshold, and then extrapolating this polynomial through the excluded region.

Finally, we use N to represent the total number of loans across certain DTI ranges. N is introduced to normalize loan quantities and compute distribution fractions. The same subscripts (h, l) and superscripts ($pre, post$) apply. For example, N_l^{post} stands for the total number of low-credit-score loans extended in the post-event period. $\hat{N}_l^{post}$ denotes the corresponding, counterfactual number.

With the above notations, we lay out the following assumptions underlying the bunching estimation.

Assumption 1. *The market for high credit score borrowers (i.e., $FICO > 620$) is not affected by the policy change.*

$$\hat{n}_{hd}^{post} = n_{hd}^{post} \quad (1)$$

Assumption 2. *There exists a maximum DTI bin $\bar{d}$ such that the total volume of low-credit-score loans with $DTI \leq \bar{d}$ is unaffected by the policy.*

$$\sum_{d=0}^{\bar{d}} \hat{n}_{ld}^{post} = \sum_{d=0}^{\bar{d}} n_{ld}^{post} \triangleq N_{l\bar{d}}^{post} \quad (2)$$

$N_{l\bar{d}}^{post}$ denotes the observed total number of low-credit-score loans with DTI below $\bar{d}$ extended after the policy event. Assumption 2 allows us to normalize loan quantities and compare them across the low- and high-credit-score markets. This is necessary because one market is significantly larger than the other. Under this assumption, we can divide the number of loans in each DTI bin by total loan counts to the left of $\bar{d}$, since lending activities at $DTI < \bar{d}$ are not affected by the policy and can be used as the scale. As seen from Figure 2, loan volume remained relatively constant for low credit score borrowers at DTI ratios below 35. Based on this pattern, it is reasonable to set $\bar{d} = 35$. In our analysis, we also experiment with $\bar{d}$ being 32, 34, and 36 to test the robustness of our findings.

Assumption 3. *The change in the (normalized) number of low CS loans in a given DTI bin between the pre- and post-periods would have been the same as the corresponding change in the high CS market in the absence of the policy.*

$$\frac{\hat{n}_{ld}^{post}}{N_{l\bar{d}}^{post}} = \frac{n_{ld}^{pre}}{N_{l\bar{d}}^{pre}} + \left(\frac{n_{hd}^{post}}{N_{h\bar{d}}^{post}} - \frac{n_{hd}^{pre}}{N_{h\bar{d}}^{pre}} \right) \quad (3)$$

Assumption 3 is the crucial assumption that establishes our counterfactual. It states that the distribution changes in the high-credit-score market represent the counterfactual for the low-credit-score market. The

first term, $\frac{n_{ld}^{pre}}{N_{ld}^{pre}}$ is the pre-event observed distribution of loans for each DTI grid in the low-credit-score market. The second term, $\left(\frac{n_{hd}^{post}}{N_{hd}^{post}} - \frac{n_{hd}^{pre}}{N_{hd}^{pre}}\right)$ is the changes in the normalized distribution of high-credit-score loans around the policy event. By taking the sum of the two terms, we assume that absent the policy reform, the changes in the DTI distribution among low-credit-score loans would have been the same as those among high-credit-score loans.

Figure 3 plots the actual and counterfactual distributions of loans at each DTI grid for low-credit-score borrowers. The red solid line represents n_{ld} , the actual number of loans issued for each DTI grid d , and the blue dashed line represents $\hat{n}_{ld}$, the counterfactual number of loans based on Assumption 3. We first notice a clear bunching of loans right below the $DTI = 43$ threshold in the counterfactual distribution. The number of loans spikes at 43, and drops at 44. Such a bunching pattern is visibly smaller in the actual, post-policy distribution. This contrast suggests that the requirement for human underwriting for low-DTI borrowers leads to the bunching of loans under the $DTI = 43$ threshold. Furthermore, Figure 3 shows a large increase in the number of loans above the $DTI = 43$ threshold post-policy of over 100% for most DTI bins. The significant magnitude of this expansion is consistent with our reduced-form evidence in Figure 2 and Figure B.2.

FIGURE 3 ABOUT HERE

One concern with the above pattern is that we might be capturing a general trend of loosening lending standards towards highly levered, low-credit-score borrowers over time. If this is the case, we should observe the same pattern at a different point in time. We thus provide a placebo analysis in Internet Appendix Figure B.3 where we choose various other pseudo events, including August of 2014, 2015, 2018, and 2019. The event windows surrounding those dates are outside of the post-event window of the true policy shock. Accordingly, we observe the bunching of loans at $DTI = 43$ both in the counterfactual and actual distributions, with no significant difference between the two around the pseudo event. This means that the reduction of bunching in Figure 3 is unlikely due to a general time trend, but instead directly related to the FHA policy.

Quantifying Credit Expansion

Under Assumptions 1 through 3, we quantify the changes in loan volume due to the FHA policy change regarding underwriting procedures. Our main focus is to identify the overall increase in credit above the unaffected DTI region, i.e., $DTI \geq \bar{d}$, also referred to as the “extensive margin” effect. Formally, it is defined as the fraction of loans granted to borrowers who would otherwise not have applied or been approved without

the policy (i.e., counterfactual scenario):

$$\Delta \text{Loans Originated} = \frac{1}{\hat{N}_l^{post}} \sum_{d=\bar{d}}^{50} (n_{ld}^{post} - \hat{n}_{ld}^{post}) \quad (4)$$

The expression inside the parentheses indicates the additional number of low-credit-score loans with DTI above $\bar{d}$ due to the policy change. $\hat{N}_l^{post}$ denotes the counterfactual total number of low-credit-score loans extended after the policy event and is used as a scaling factor. The DTI variable is winsorized at the 1st and 99th percentiles and rounded to the nearest integer. To obtain confidence intervals, we use a block bootstrap procedure, drawing 1000 random samples clustered by origination month and re-estimating the parameter at each iteration. The origination-month clustering helps mitigate concerns about temporal correlation in the error term. For example, macroeconomic shocks (e.g., changes in unemployment, interest rates) may affect loan originations for all borrowers in a given month, generating correlation across DTI bins within that month.

In addition, we provide two auxiliary measures of the change in credit quantity, directly computing the fractional growth in the number of loans for a specific DTI range relative to the counterfactual. We compute such growth rates separately for low-DTI ($\bar{d}$ to 43) and high-DTI (43 to 50) ranges.

$$g(\text{Volume}) (\text{high DTI}) = \frac{\sum_{d=43}^{50} (n_{ld}^{post} - \hat{n}_{ld}^{post})}{\sum_{d=43}^{50} \hat{n}_{ld}^{post}}$$

$$g(\text{Volume}) (\text{low DTI}) = \frac{\sum_{d=\bar{d}}^{43} (n_{ld}^{post} - \hat{n}_{ld}^{post})}{\sum_{d=\bar{d}}^{43} \hat{n}_{ld}^{post}}$$

Changes in DTI Distribution

Figure 3 shows that the policy change gives rise to a drastic shift in the DTI distribution. Not only do lenders expand credit provision, attracting new borrowers to enter the market and apply for a mortgage, existing borrowers may also decide to increase loan size after the policy change, increasing their DTI ratio from below to above 43. We label the latter effect as the “intensive margin” effect, and seek to quantify it in this section.

Following DeFusco et al. (2020), we measure the reduction in volume in range $\bar{d} \leq DTI \leq 43$ around the policy change. Again, we compare the fraction of loans in this range relative to the counterfactual scenario:

$$\Delta \text{Low DTI Loans} = \frac{1}{\hat{N}_l^{post}} \sum_{d=\bar{d}}^{43} (n_{ld}^{post} - \hat{n}_{ld}^{post}) \quad (5)$$

In the parentheses, $n_{ld}^{post} - \hat{n}_{ld}^{post}$ indicates the change in low-DTI loans compared to the counterfactual case without the policy at DTI d . We focus on the DTI ranging between $\bar{d}$ to the threshold 43 because below $\bar{d}$, loan quantity remains unaffected by the policy (Assumption 2).

Finally, we analyze the change in the average DTI ratio of approved loans, defined below:

$$\Delta Average DTI = \sum_{d=1}^{50} d \left(\frac{n_{ld}^{post}}{N_l^{post}} - \frac{\hat{n}_{ld}^{post}}{\hat{N}_l^{post}} \right) \quad (6)$$

Results

Results are reported in Table 2. In column (1), we set the cap for the unaffected DTI threshold $\bar{d}$ to be 35, following the pattern displayed in Figure 3. Results from the extensive margin ($\Delta Loans Originated$) suggest a significant increase by 6.5% among low-FICO borrowers. Relative to the 12 months prior to the policy implementation, the loan volume in the high-DTI range (43–50) grows by 122%.

TABLE 2 ABOUT HERE

As indicated by Figure 3, some of the credit volume growth in the high-DTI range may arise from households increasing their leverage ratio, crossing the DTI 43 threshold. Consistent with this conjecture, we observe a 16.1% decline in low-DTI loan volume compared to pre-event levels, or a 9.3% decline relative to the counterfactual benchmark. Overall, such a shift leads to a sizable increase in the average DTI ratio of the low-FICO market by 1.1 percentage points.

In columns (2) through (4) of Table 2, we alternate $\bar{d}$ to be 32, 34, and 36. Effects remain highly statistically significant and stable in magnitude.

How do our bunching estimates square with the descriptive patterns shown in Figure 1? We provide a back-of-the-envelope calculation to reconcile the quantities. Denote x as the percentage growth of high-DTI loans among low-credit-score borrowers that is consistent with the pattern in Figure 1. Also denote $HighDTIShare_{pre}$ ($HighDTIShare_{post}$) as the percentage of high-DTI loans among low-credit-score borrowers prior to (after) the policy. The share of high-DTI loans should evolve around the FHA policy as:

$$HighDTIShare_{post} = (HighDTIShare_{pre} + \Delta Low DTI Loans + x) / (1 + x)$$

In other words, the post-policy high-DTI share should be the sum of three components: (1) existing borrowers, (2) borrowers that shift from the low- to the high-DTI segment, and (3) borrowers that newly enter the market. The underlying data of Figure 1 suggest that $HighDTIShare_{pre} = 7.7\%$ and $HighDTIShare_{post} = 26.9\%$. Also given that $\Delta Low DTI Loans = 9.3\%$ by Table 2, we get $x = 13.5\%$.

This is the expansion of high-DTI, low-FICO borrowers inferred based on the patterns shown in Figure 1. Applying the same formula for the high-FICO markets, we have $x = 5.3\%$.¹⁵ Thus, the differential change in high-DTI loans between low- and high-FICO markets is $8.2\% = 13.5\% - 5.3\%$, close to our bunching estimate of 6.5%.

We further verify that the parallel pre-trend assumption holds. In Figure 4, we present the $\Delta Loans$ *Originated* and the $g(Volume)$ (*high DTI*) for each quarter during the [-4, +4]-quarter event window. The base period for each estimate is the second quarter of 2016, the calendar quarter prior to the policy adoption. The third quarter of 2016 is omitted because it contains both pre- and post-event periods. The pattern suggests that the expansion in high-DTI credit did not exist prior to the policy change, but is statistically significant after the policy, which helps alleviate concerns that our quantity effects might be driven by pre-existing trends.

FIGURE 4 ABOUT HERE

Taken together, results from our estimations suggest that the FHA policy leads to a substantial increase in the origination of loans to low FICO borrowers along with an increase in the average loan DTI. The increase in DTI is driven by both borrowers switching from low- to high-DTI loans and the entry of new borrowers.

We caution the interpretation of the intensive margin metrics for two reasons. First, the policy change may encourage households to enter the housing market and search for mortgage deals. The search may end up with households purchasing properties of moderate value, leading to a DTI ratio below 43. This means that $\Delta Low\ DTI\ Loans$ does not only account for low-DTI households shifting to the high-DTI market, but the net effect from the exit of the low-DTI households and the entrance of new buyers. Second, borrowers may have some room for discretion when reporting their income around the DTI threshold, such as whether to include certain bonus income. Prior to the policy reform, borrowers may have had a greater incentive to boost their income, so that their DTI ratio stays under the 43 threshold. This could lead us to overestimate the intensive margin effects. Given these considerations, we focus on the overall increase in loan volume (i.e., the extensive margin, $\Delta Loans$ *Originated*) for the remainder of the analysis, since that metric is less subject to noise and provides a more straightforward proxy of credit expansion. In addition, we present $g(Volume)$ (*High DTI*), which measures the fractional increase in high-DTI loan volume, and helps us reconcile the magnitude of the credit volume expansion with descriptive evidence.

¹⁵Here, the underlying data for Figure 1 show that $s_{high\ DTI, pre} = 34\%$ and $s_{high\ DTI, post} = 37.3\%$. Given that this market segment is not affected by the FHA policy, we assume zero shift ($a = 0$) in this market.

5 Delinquency

Does the policy-induced credit expansion for low-credit-score, high-leverage borrowers engender greater risk exposure for lenders and the FHA? We seek to answer this question by examining how loan delinquency changes around the FHA policy reform using two research designs. First, we estimate a difference-in-difference regression for high-DTI and low-DTI loans separately:

$$Delinquent_{i,t} = \beta Treated \times Post + \tau_t + \phi_f + \epsilon, \quad (7)$$

where i denotes loans, t denotes origination month, and f denotes FICO. $Post$ is an indicator for months after the policy change (August 2016). $Treated$ is an indicator that equals one if the borrower’s credit score is below 620, and zero otherwise. Our coefficient of interest is β , which indicates the change in delinquency among the low-credit-score loans relative to the high-credit-score ones. We add controls, fixed effects, and interactive fixed effects in stages. Given that the FHA policy has little impact on borrowers with DTI under 35, we restrict the testing sample to loans with DTI ratios above $\bar{d} = 35$. Similar to the quantity estimation, we cap the DTI ratio at 50 since there were no loans with DTI above 50 and credit scores below 620 prior to the policy.¹⁶ For the main specification, we track whether the borrower incurs delinquency over the next two years for each loan, following [Bosshardt et al. \(Forthcoming\)](#). We perform robustness tests using different time horizons to measure delinquency.

Results are reported in Panel A of Table 3. Columns (1) through (3) present results for high-DTI loans; while columns (4) through (6) report results for low-DTI loans. For each sample, we start with a relatively sparse specification (columns (1) and (4)), and only impose origination month and FICO fixed effects. Origination month fixed effects help remove macro-level changes in lending standards, while FICO fixed effects allow us to compare loans of similar credit scores. In the next specification (columns (2) and (5)), we include origination month-by-DTI fixed effects to account for the time-varying ability to repay for households with certain levels of leverage, and FICO-by-DTI fixed effects to compare loans of similar credit scores and leverage ratios. The controls include the natural logarithm of loan amount and borrower application income. In the last specification (columns (3) and (6)), we add county and lender fixed effects to control for time-invariant county and lender characteristics that may affect default rates, including the selection into certain counties and changes in lender composition.

TABLE 3 ABOUT HERE

¹⁶In Section B of the Internet Appendix, we present a comparison of the delinquency rates between loans with DTI 43–50 and DTI above 50. We find that conditional on observables, delinquency rates are higher below 50 than above 50, suggesting that algorithms also significantly expanded credit supply in the DTI>50 region while controlling risk, consistent with our main results.

Across all specifications, the coefficients on $Treated \times Post$ are economically small and statistically insignificant for both high- and low-DTI loans. This result suggests that the policy change does not increase the default rate of low-credit-score borrowers relative to high-credit-score ones. If anything, the coefficient estimates are mostly negative, suggesting a small drop in the average delinquency rate, conditional on observables.

We next estimate triple-difference regressions, comparing the changes in delinquency rates for treated borrowers between high- and low-DTI loans:

$$\begin{aligned} Delinquent_{i,t} = & \gamma_1 Treated \times High\ DTI \times Post + \gamma_2 Treated \times High\ DTI \\ & + \gamma_3 Treated \times Post + \gamma_4 High\ DTI \times Post + \tau_t + \phi_f + \epsilon, \quad (8) \end{aligned}$$

where *High DTI* is a dummy variable that equals one if the DTI ratio is above 43, and zero otherwise. Panel B of Table 3 reports the results. Again, there is no statistically significant difference in the changes in delinquency rates between the two subsamples, with the coefficient estimates on the triple interactions being negative and small.

While Table 3, column (3) of Panel A shows a decline in delinquency rates of -0.79 percentage points, the upper bound of the one-sided 95% confidence interval allows for an increase of up to 1.36 percentage points. Thus, we cannot statistically rule out the possibility that the FHA policy modestly raises delinquency rates. However, even this worst-case increase would be economically small when viewed against three relevant benchmarks. First, it is small relative to the pre-policy average delinquency rate of 16.31 percentage points for low-credit-score, high-DTI borrowers. Second, it is modest compared to the cross-state standard deviation of delinquency rates for this group, which is 11.29 percentage points. Third, it is well below FHA's own threshold for an "acceptable risk" delinquency rate of 22.4 percentage points (McFarlane, 2010).

To further gauge the risk of borrowers added by the policy, we perform a back-of-the-envelope calculation to infer the delinquency rate of marginal borrowers, who would be approved under algorithmic underwriting but rejected under human underwriting, relative to inframarginal (existing) borrowers. We estimate that the 95% upper-bound increase in overall delinquency rate is 1.36 percentage points, which reflects (1) the change in the delinquency of inframarginal borrowers and (2) the increased presence of marginal borrowers with higher delinquency rates than inframarginal ones. Suppose that the delinquency rate of inframarginal borrowers remains unchanged after the policy. Given that marginal borrowers represent a 121.5% increase in the low-credit-score high-DTI FHA market, our estimates suggest that the upper bound of the differential delinquency between marginal and inframarginal borrowers is 2.46 percentage points (i.e., $\frac{2.46 \times 1.215 + 0 \times 1}{1.215 + 1} = 1.36$). A parallel calculation would suggest the point estimate of the delinquency differential

to be -1.44 percentage points.¹⁷ Both of these changes are small in magnitude compared to the cross-state standard deviation in delinquency rates of 11.29 percentage points.

One concern with our delinquency analysis is that our test may not have sufficient power to detect changes in delinquency resulting from the policy due to the time period under consideration. For example, one may argue that delinquency rates were low for mortgages originated around 2015–2017, because house prices and economic conditions were stable or improving during that period. Counter to this argument, we note that the average 2-year delinquency rate in our sample is not negligible, but hovers around 6% overall, 13% for low-credit-score borrowers, and 16% for low-credit-score, high-DTI borrowers. To further address this concern, we conduct an additional analysis in Table 4, where we separately examine the effect of the policy on delinquency rates across locations with different unemployment growth rates. To the extent that negative life events are the primary driver of mortgage defaults (Ganong and Noel, 2023), the above concern would suggest that the FHA policy change should induce higher delinquency rates in areas with the highest unemployment growth. However, we do not find this to be the case. Even in counties that experience the highest increase in unemployment rate, we continue to see muted effects of the policy on delinquency rates. If anything, delinquency rates decline for the treated group in those counties.

TABLE 4 ABOUT HERE

We provide evidence supporting the parallel-trend assumption for the delinquency tests. To do so, we perform dynamic triple-difference analysis and analyze the differential changes in delinquency for highly levered, low-credit-score borrowers around the policy date. Figure 5 reports the results. The dots represent the point estimates of the triple-difference coefficients, while the vertical lines represent confidence intervals. We do not observe significant pre-event trends.

FIGURE 5 ABOUT HERE

In Figure 6, we report the changes in delinquency rates around the policy event with different local economic conditions, measured by county unemployment growth rates. Panel A (D) reports the changes in delinquency in counties with the bottom (top) quartile of unemployment growth. Suppose a heavier reliance on algorithmic underwriting admitted more “fragile” borrowers who are prone to default during poor economic conditions. In that case, we should observe an increase in delinquency rate in areas with greater unemployment rate growths. However, we do not find that to be the case. Delinquency rates remain unchanged across counties with better and worse economic conditions.

¹⁷This can be solved via the equation $\frac{x \times 1.215 + 0 \times 1}{1.215 + 1} = y$, where $y = -0.79\%$, with -0.79% being the point estimate in column (3) of Panel A in Table 3, and x represents the point estimate of the delinquency difference between marginal and inframarginal borrowers, which is the unknown variable that we seek to solve.

FIGURE 6 ABOUT HERE

We provide multiple additional analyses to test the robustness of our findings and address remaining concerns regarding delinquency rates. First, it is possible that we do not find any significant effects on delinquency rates in the limited time horizon that we focus on. We address this concern in Table C.1, where we look at 3-year, 4-year, and 5-year delinquency rates. Our inferences remain unchanged. We note that the 4-year and 5-year delinquency rates include the COVID-19 forbearance period, when delinquency increased substantially due to the ability of borrowers to take on forbearance, and our results suggest that the uptake of this forbearance has not differentially increased for low-FICO, high-DTI borrowers post-policy. Second, it is possible that our measure of delinquency, which focuses on 90-day delayed payment, does not capture milder levels of borrower distress. In Table C.2, we evaluate less severe delinquencies, including 30-day and 60-day delinquencies, and continue to find no changes associated with the FHA policy reform.

A natural explanation for not finding a significant delinquency effect is that the additional information collected by human underwriters has limited incremental value relative to the hard information used by the FHA's underwriting algorithm. We discuss and rule out several alternative explanations. First, it is possible that human underwriters do gather valuable information, but loan officers are able to push against their recommendations due to volume incentives, making human underwriting less effective. This channel seems unlikely given the significant concerns voiced by lenders regarding FHA scrutiny over delinquency during our sample period (Frame et al., Forthcoming). Second, human underwriters' role can be weakened if lenders purposefully extended loans to high-risk borrowers to comply with the Community Reinvestment Act (CRA), despite the recommendations of human underwriters. This should not be the main driver of our results for two reasons. First, the majority of the high-risk loans in our sample are made by non-bank lenders who are not subject to such CRA requirements. Second, we continue to find no change in delinquency rates when excluding CRA census tracts from the sample (Table C.3). Finally, it is possible that borrowers in this market exhibit homogeneous risk profile, leaving little to no role for underwriters, either humans or algorithms. Contrasting this argument, Berg (2015) documents a prominent role for risk managers.

We note that, while we do not detect increases in delinquency rates within each DTI bin, there is a significant increase in the number of low-credit-score, high-DTI loans after the policy change, which could lead to an increase in overall, unconditional delinquency rates. In Section 9, we perform a back-of-the-envelope calculation to quantify how the policy may shape the FHA's insurance premiums.

6 Heterogeneous Effects Across Race and Income

In this section, we analyze the heterogeneous effects of the policy across various borrower types, including income, race, and ethnicity. This analysis sheds light on the ongoing discussion regarding the distributional impact of algorithmic underwriting.

We begin by exploring whether the effects differ across borrower race, motivated by persistent academic and policy discussions on racial disparities in mortgage markets. Recent research suggests that access to credit and credit terms differ systematically by race even after conditioning on observables (Bhutta and Hizmo, 2021; Bartlett et al., 2022). Moreover, algorithms can produce racial disparities even though race is not used as an underwriting variable (Fuster et al., 2022). The focus on borrower income is motivated by important policy considerations to provide financing to low-income households.¹⁸ Does a shift towards algorithmic underwriting mitigate or aggravate racial and income inequalities? We seek to answer this question by separately analyzing the impacts of the FHA underwriting policy for Black and non-Black borrowers, as well as lower- and higher-income borrowers.

For each borrower subsample, we examine the credit expansion as well as changes in loan delinquency rates. Credit expansion is measured both using the growth in low-credit-score loans ($\Delta \text{Loans Originated}$) and low-credit-score, high-DTI loans ($g(\text{Volume Growth}) (\text{high DTI})$) relative to their counterfactual levels. To measure changes in delinquency risk, we report the estimated β s (i.e., on $Treated \times Post$) from the delinquency regression Equation (7) for high- and low-DTI loans separately. These coefficients capture the differential change in delinquency rates for a DTI range between low- and high-credit-score borrowers following the FHA policy. For all of these estimates, we also present the differences between race and income subsamples and their corresponding statistical significance.

Results are reported in Table 5. In Panel A, we examine the differential policy impacts across borrower race. The policy-induced credit expansion is largely concentrated among non-Black borrowers, with magnitudes around 7.3%, while essentially non-existent among Black borrowers. In terms of the percentage volume growth, non-Black borrowers experience a 126% increase in low-DTI loans relative to the counterfactual level, while Black borrowers experience significantly lower growth, at around 86%.

TABLE 5 ABOUT HERE

Next, in Panel B, we compare the policy effects for higher- and lower-income borrowers. Income subsamples are defined by whether the ratio of borrowers' household income scaled by the MSA median income (i.e., adjusted income) lies above the sample median. This location-based adjustment helps remove

¹⁸For instance, under the Community Reinvestment Act (CRA), banks need to meet the credit needs of low- to moderate-income neighborhoods; HMDA requires loan-level reporting on borrower and tract income so regulators and the public can assess whether lenders are serving community housing needs and to detect disparate patterns; The FHFA sets benchmarks for low- and very-low-income lending that Fannie Mae and Freddie Mac must meet.

the heterogeneity of income created by cross-region differences in economic conditions and lending standards. We find that the increase in loan volume is relatively stronger for higher-income borrowers than lower-income ones. Borrowers with above-median adjusted income experience a 6.7% increase in the overall loan volume after the policy change, or a 160% relative increase in high-DTI loans relative to the pre-policy period. Those growth rates are 4.5% and 78% for lower-income borrowers.

Given that Black borrowers have relatively lower income than others, the heterogeneity across races may simply reflect their income differential. To address this concern, we construct a propensity score weighted sample, where non-Black borrowers are weighted based on their income and FICO score bins to match those of Black borrowers, based on the covariance-balanced propensity score weighting approach. The details of the approach are outlined in Internet Appendix Section A. Results from this weighted sample are reported in Panel C. We again find the policy-induced credit expansion to be concentrated on non-Black borrowers, in contrast to a near-zero point estimate for Black borrowers. In Internet Appendix Section B, we employ an alternative weighting approach which delivers similar estimates. Matching race subsamples by income does not materially affect our inferences, likely because the FHA program is designed to expand access to homeownership for low-to-moderate-income borrowers, resulting in similar income distributions among Black and non-Black borrowers in this market. This finding highlights the challenges of using automation to promote financial inclusion in historically disadvantaged populations.

One might argue that concerns of being accused of unequal lending standards may make lenders relax human underwriting criteria for Black and low-income borrowers prior to the policy. This could explain the weaker credit expansion among Black and lower-income borrowers. We test this explanation by examining the changes in delinquency rates and observed credit quality of Black and lower income borrowers post-policy. If human underwriters had imposed laxer criteria on Black and lower-income borrowers than on other borrowers before the policy, the reduction of human input should lead to a relative improvement in credit quality for Black and low-income borrowers compared to other borrowers. Measuring credit quality using ex post loan delinquency rate and credit scores at application, we do not find evidence consistent with this prediction. Table 5 shows no relative decreases in delinquency rates for Black and low-income borrowers. Table D.3 of the Internet Appendix suggests no significant differential increase in the average FICO scores of Black relative to non-Black borrowers, or low- relative to high-income borrowers. If anything, non-Black borrowers exhibit a significantly greater increase in FICO scores than Black borrowers.

In Figure D.1, we provide the bunching subsample graphs showing the heterogeneous credit expansions across borrower race and income. In Table D.4 of the Internet Appendix, we show that Hispanic borrowers experience a similar credit expansion to non-Hispanic borrowers after the policy, but exhibit a greater decline in delinquency rates than non-Hispanic borrowers. One possible explanation for this differential delinquency effect is that algorithms may overcome human frictions such as language barriers or communication hurdles,

and potentially are able to better screen Hispanic borrower quality than human underwriters.¹⁹

7 Economic Mechanisms

Our baseline results suggest that a heavier reliance on algorithm underwriting can expand credit supply without a significant increase in delinquency rate. We find evidence consistent with a few underlying channels. First, algorithmic underwriting can mitigate lenders’ concerns regarding FHA scrutiny (a “regulatory concerns” channel). Given that the algorithms are endorsed by the FHA, relying more on the algorithm can reduce lenders’ concerns that the FHA may deny the insurance claims they submit based on human underwriters due to deficient underwriting, and may also reduce litigation risk surrounding humans’ underwriting practices. Second, using algorithms can reduce the information asymmetry regarding human underwriting quality among financial intermediaries. To the extent that algorithms have predictable stringency, aggregators face less uncertainty regarding the quality of the loans they purchase, thus providing more liquidity to lenders. Third, algorithms may place greater weight on standardized, structured borrower information, such as credit histories and verifiable income, leading to larger credit expansions in settings where such information is more readily available. We also rule out other explanations, including a “capacity constraint” channel, which suggests that using algorithms allows lenders to meet credit demand when they have limited time, energy, and human capital, and a lender-entry channel suggesting that the policy increased lender entry due to the reduction in underwriting costs. We discuss these mechanisms and our evidence in turn.

7.1 Regulatory Concerns

We evaluate the regulatory concerns channel by examining the differential effects of the FHA underwriting policy on lenders facing varying degrees of regulatory concerns. First, we compare bank and nonbank lenders. To the extent that nonbank lenders face less stringent regulatory scrutiny and can securitize a greater share of their loan portfolio (Benson et al., 2023b), they should be less concerned about regulatory risk compared to bank lenders. If the reliance on algorithmic underwriting helps lenders overcome regulatory concerns, we expect such effects to be more pronounced among banks than nonbanks.

We compare the expansion of high-DTI, low-credit-score loans for bank and non-bank lenders, whereby “lenders” are defined either by originators or issuers. Originators interact more closely with the borrower, whereas issuers sell the originated loan to Ginnie Mae. Both make credit decisions and are potentially liable for the loans they originate (Benson et al., 2023b). Results are presented in Panels A and B of Table 6.

¹⁹Liu (2023) finds that language frictions are a significant hurdle to Spanish-speaking borrowers, and that a FHFA policy reducing language frictions had a negative though statistically insignificant effect on the delinquency rates of the treated group post-policy.

Consistent with our conjecture, bank lenders increase credit provision significantly more after the FHA policy change than non-bank lenders, consistent with algorithmic underwriting alleviating regulatory concerns.

TABLE 6 ABOUT HERE

Second, we isolate bank lenders (either issuers or originators) that faced litigation regarding their practice in the FHA markets prior to the policy change, and compare how their lending quantity changes around the policy relative to non-litigated banks. During the 2010s, the Department of Justice (DoJ) launched a series of lawsuits against large lenders, accusing them of fraudulent practices in qualifying borrowers for the FHA insurance (Frame et al., Forthcoming). These lawsuits significantly reduced credit provision by the litigated lenders. We define “litigated lenders” as bank lenders that have settled their FHA-fraud litigations before the pre-policy window, based on the list from Frame et al. (Forthcoming). We conjecture that litigated banks faced greater regulatory concerns prior to the FHA policy than non-litigated banks and likely withheld lending to high-risk borrowers. The policy change should reduce the concerns among litigated lenders and increase their issuance volume. Results in Panel C lends support to this conjecture. Litigated banks increase their low-credit-score loans in the FHA markets by 21.2%, a significantly greater expansion compared to the 9.2% by non-litigated banks. The difference in the credit expansion provided by litigated and non-litigated banks is around 12 percentage points, which is economically meaningful but has relatively low statistical significance ($p < 0.1$). This is likely due to the limited statistical power, given that prior to our event, only 8 banks faced lawsuits regarding their FHA insurance practices.

Figure D.2 presents graphical evidence regarding the changes in DTI distribution for the subsample of loans extended by banks, nonbanks, litigated banks, and nonlitigated banks. The patterns are consistent with the results shown in Table 6.

7.2 Information Frictions Among Intermediaries

In the U.S., many small mortgage lenders rely on aggregators to securitize the loans they originate, which provides crucial liquidity to facilitate further loan origination. We refer to aggregator-dependent originators as indirect sellers. While aggregators do not conduct the underwriting procedures in-house, they are legally liable for delinquencies of loans they securitize (Benson et al., 2023a,b). They are also more likely to be servicers of the loans they securitize, and thus bear elevated servicing costs in the event of FHA delinquencies (Benson et al., 2023b). Given that aggregators cannot fully observe or control the underwriting quality of indirect sellers, they may be wary of errors or conflicting incentives of human underwriters employed by those sellers. As the FHA underwriting policy reform gives more decision power to algorithms, which have uniform underwriting stringency, it may alleviate aggregators’ concerns regarding unobservable human

underwriting quality. We thus expect the policy to encourage purchase by aggregators and in turn, the credit provision by indirect sellers. Results in Panel A of Table 7 confirm this prediction. After the FHA policy adoption, indirect sellers increase FHA credit by around 12 percentage points, a magnitude significantly larger than the growth by direct originators. Figure D.3 provides visual evidence regarding the differential credit expansion between indirect sellers and other originators.

TABLE 7 ABOUT HERE

One potential concern with this result is that indirect sellers are more likely to be smaller non-bank institutions and the difference between indirect sellers and other originators could be driven by lender type and bank size. We address this concern in Panel B, showing that the results persist when we assign weights to observations associated with indirect sellers, so that the lender type (bank vs. nonbank) and size distributions among those observations match those from direct sellers. Details of the weighting procedure are discussed in Internet Appendix D.

7.3 Availability of Standardized Data

Next, we explore how the availability of standardized data helps explain the difference between algorithmic and human underwriting. We hypothesize that AUS may either have an advantage in processing standardized borrower data or place greater weight on such information compared to human underwriters. If this is at play, the transition to AUS should shift mortgage approval decisions toward borrowers with more standardized, hard information, such as payment histories and documented income.

We test this prediction in Table 8. We first compare first-time and repeat home buyers, where repeat buyers tend to have richer credit histories, especially mortgage payment records, which are standardized data that algorithms can effectively evaluate. We then compare sole mortgage applicants and borrowers with co-applicants, where co-applicants provide more than one set of verified incomes and credit profiles. Finally, we compare areas with above- and below-median rates of self-employment, since high self-employment areas may have higher shares of irregular income that are not well-captured using standardized data and may require more human information collection and input. Consistent with the mechanism proposed above, we find significantly greater credit expansions for repeat home buyers, loans with co-applicants, and low self-employment areas. The difference between each pair of subsamples is generally statistically significant. Figure D.4 of the Internet Appendix depicts the differential credit expansion across the borrower characteristics discussed above.

TABLE 8 ABOUT HERE

Accounting for the possibility that the above characteristics may be correlated with race and income, we balance the racial profile and income levels between each pair of subsamples by assigning weights to each observation according to a propensity-score weighting approach (explained in Internet Appendix E). Results remain virtually unchanged.

Together, our results suggest that AUS leads to greater credit expansion among borrowers with more standardized income and repayment data. Our evidence highlights the limitations of AUS in extending credit to those with more complex, unconventional financial profiles within the high-risk segment.

7.4 Capacity Constraints

We assess the capacity constraints channel by testing whether the policy affects credit quantity differently across lenders facing more and less capacity constraints. We gauge lenders' capacity constraints in two ways. First, we track the seasonality of mortgage applications and classify the six months with the highest application volumes as the "high season" and the remaining six months as the "low season," following the method introduced in [Jo and Liu \(2024\)](#). The high season includes March through August, and the low season includes September through February. Second, we measure lender capacity constraints using the year-on-year growth in loan application volume for each lender-zip code. To the extent that lenders cannot quickly adjust the number of underwriters in response to changing demand ([Fuster et al., Forthcoming](#)), underwriters likely face time and attention constraints in markets or periods with rapid application growth, reducing their ability to meet demand. Such capacity constraints should be alleviated when underwriting can be delegated to algorithms after the FHA policy. Thus, we expect capacity-constrained lenders to expand credit supply more compared to unconstrained lenders after the policy shock.

To test this conjecture, we separately examine the credit expansion for lenders facing higher or lower capacity constraints, i.e., lenders issuing loans in the high season or the low season, in markets with high or low application growth. Results are reported in Table E.1 in the Internet Appendix. We find that after the FHA policy shock, both capacity-constrained and unconstrained lenders are able to issue more loans, but the quantity expansion is not statistically significant between groups.

In Figure D.5 of the Internet Appendix, we present sub-sample graphs on the empirical and counterfactual DIT distributions for lenders facing higher and lower capacity constraints. The visual evidence suggests little difference across subsamples. We perform robustness tests for this result in Table E.2, where we show that results are robust when we measure local market by census tract rather than by zip code, and when we measure congestion using non-FHA applications. Collectively, our evidence does not provide support for the capacity constraints channel.

7.5 Lender Entry

Finally, we evaluate the extent to which our credit expansion results are driven by lender entry into the market. As shown in Table E.3, 81% of low-FICO, high-DTI loans are originated by existing lenders post-policy, and 19% by new lenders. Recall that Figure 2 shows that the quantity of low-FICO, high-DTI loans expanded over 100% in the post-policy period. This means that lender entry is unlikely to be the main driver of the policy-induced credit expansion.

8 Implication for Borrowers

In this section, we design two analyses to investigate the potential impact of the FHA policy on borrowers. First, we look into the changes in borrowing costs for high-leverage borrowers. Second, we track households' location choices and examine whether the credit policy allows them to migrate to higher-quality neighborhoods.

8.1 Loan Pricing

Is the expansion of FHA credit associated with higher interest charges? We seek to address this question by examining the changes in the interest rate spreads charged on high- and low-DTI loans around the policy shock, and compare such changes between low- and high-credit-score borrowers. Table 9 reports the results. The format of this table follows closely that of the delinquency analysis. From Panel A, we find no changes in interest rate spreads among high-DTI loans, but a statistically significant increase for low-DTI loans in some specifications. This might be caused by changes in borrower characteristics among the low-DTI borrowers. In Panel B, we confirm that interest rates increase less for treated borrowers in the low-DTI sample relative to the high-DTI sample. The coefficient of $Treated \times Post \times High\ DTI$ suggests that the differential change in interest rates for highly levered, low-credit-score borrowers is relatively small, around 3 basis points. In Figure 7, we do not observe any pre-policy changes in interest rates, confirming that the previous findings are not driven by pre-existing trends.

TABLE 9 AND FIGURE 7 ABOUT HERE

8.2 Mortgage Access and Neighborhood Choice

Recent evidence suggests that neighborhood quality varies widely across regions and significantly shapes economic opportunity, with higher-opportunity areas yielding better long-term outcomes for children (Chetty

et al., 2016, 2025). However, households often face barriers, such as information frictions, search costs, and credit constraints, that limit their ability to move (Bergman et al., 2024). In this section, we investigate the impact of increased mortgage access stemming from changes to lender underwriting regulations on individuals' subsequent neighborhood choices. Our main measures of neighborhood quality reflect neighborhoods' inter-generational upward mobility following Chetty et al. (2025), which describes the future household income ranks of children raised by parents at the 25th percentile of the national income distribution. We additionally examine the future household income ranks of children whose parents' income rank at the 50th percentile, which is more representative of the typical FHA borrowers. As shown in Table 1, the average borrower in our sample has an adjusted income that is approximately one, suggesting that their income is close to the median income of their residential area. In auxiliary tests, we utilize other location quality measures, such as school district quality and other local economic and demographic characteristics that predict upward mobility in Chetty et al. (2025).

For this analysis, we rely on the credit bureau data, which is an individual-year panel that allows us to track how people's addresses change over time. We compute the year-on-year change in one's neighborhood quality and examine whether the implementation of the FHA policy and the subsequent change in mortgage access enable her to move to better-quality neighborhoods.

Our main empirical approach is to compare individuals with a credit score below 620 (treated group) and above 620 (control group) in 2015, the year before the policy implementation. Credit score is measured by VantageScore in the Experian data. We conduct a two-stage-least-square (2SLS) analysis where the outcome variable for the first stage is *New Purchase FHA*, an indicator for whether an individual obtains a new FHA mortgage in a given year (excluding refinancing). Then, in the second stage, we further link the changes in neighborhood quality to the predicted likelihood of a new FHA purchase. Specifically, we estimate the following regressions:

$$New\ Purchase\ FHA_{i,t} = \beta_1 Treated_i \times Post_t + X_{i,t} + \alpha_{cs} + \tau_{z,t} + \phi_{g,t} + \epsilon_{i,t} \quad (9)$$

$$d(Neighborhood\ Quality)_{i,t} = \gamma_1 \widehat{New\ FHA\ Mortgage}_{i,t} + X_{i,t} + \alpha_{cs} + \tau_{z,t} + \phi_{g,t} + \nu_{i,t}, \quad (10)$$

where i indexes individuals, t indexes years, cs denotes integer credit score bins, z represents the individual's current zip code, and g captures individual characteristics such as gender, marital status, and age group (18–35, 35–45, and 45+). In our initial specification, we control for credit score fixed effects, individual characteristics, and year-by-zip-code fixed effects, the latter accounting for granular, time-varying local economic conditions. In a stricter specification, we further interact year fixed effects with individual characteristics to capture potential time-varying determinants of upward mobility related to gender, marital status, and age.

To maintain a close comparison between human and algorithmic underwriting, we further exclude the

DTI>50 borrowers in our sample to the best of our ability. Given that Experian does not provide the DTI ratios of originated mortgages, we seek to obtain this information from the Ginnie Mae-Endorsement data. Each year, we merge individuals in Experian who obtain a new FHA mortgage with mortgage records in the Ginnie Mae-Endorsement data following procedures described in Internet Appendix A.3, and drop matched borrowers who have ever obtained a new FHA mortgage with DTI>50 within our sample period. Not all borrowers can be matched, but we assume that the composition of the matched observations is representative of the true sample and weight the matched observations by the inverse of the annual match probabilities, so that the total weight of the matched observations equals the fraction of all new FHA borrowers in the Experian data. We exclude unmatched new FHA borrowers from the regression sample, and assign a weight of one to the remaining borrowers who did not get a new FHA mortgage that year.

Table 10 presents the 2SLS results. In the first stage, the treated group experiences a statistically significant increase in the likelihood of getting an FHA mortgage. The F -statistics are between 164 and 182 across different specifications, evidence of a strong instrument. In the second stage, our results suggest that greater FHA mortgage access is associated with moves to neighborhoods offering higher upward mobility. For example, columns (1)–(2) indicate that upward mobility at the 25th percentile of parents’ income improves by roughly 13–18 percentage points. Relative to unconditional means of approximately 44 percentage points, the estimated effects for compliers are economically meaningful. Columns (3)–(4) examine upward mobility at the 50th percentile and yield similar conclusions.

TABLE 10 ABOUT HERE

We further examine alternative measures of neighborhood quality, including school district ratings, which are salient information to home buyers, and additional measures identified by Chetty et al. (2025) as the strongest predictors of upward mobility, namely, average household income, poverty rates, college-educated shares, and single-parent household shares.²⁰ These variables could capture key socioeconomic conditions underpinning neighborhoods’ effect on economic opportunity. Results presented in Table F.2 of the Internet Appendix suggest an overall improvement in neighborhood quality across all dimensions. For instance, estimates in columns (1) and (2) show that school district ratings increase by approximately 1.2-1.3 units, equivalent to a shift from a 5-rated district (the sample average) to one rated between 6 and 7.

A potential concern is that our results capture pre-existing differences in migration tendencies between low- and high-credit-score individuals. To address this, we conduct a dynamic event-study analysis of the regressions in Table 10 by directly regressing the first and second stage outcomes on the interactions of *Treated* and event-year indicators. Results from the dynamic analysis are reported in Figure 8. We find

²⁰Chetty et al. (2025), Figure IV, identifies these neighborhood characteristics as having the strongest correlations with upward mobility. We select variables above a cutoff of correlation of 0.5, where there is a natural break point.

no significant pre-trends for upward mobility measures. There is a pre-event decline between 2014 and 2015 in new FHA mortgage attainment, suggesting that before the policy was implemented, low-credit-score individuals became less likely to take up new FHA mortgages compared to high-credit-score individuals. These patterns are in the opposite direction of the policy effects.

FIGURE 8 ABOUT HERE

We note several limitations to this analysis. First, because DTI is not observed until mortgage origination, our analysis necessarily relies on a difference-in-differences framework rather than a bunching framework that involves an additional comparison with low DTI borrowers. Nevertheless, the body of evidence suggests migration toward opportunity-rich neighborhoods by low-credit-score borrowers following the policy, making our findings at least *suggestive* that credit supply influences neighborhood choice. Further research on this question remains valuable. Second, this analysis may be subject to measurement error because the VantageScore in the Experian data is measured at year-end: about 8 months before the policy change, rather than immediately before, and because Experian’s VantageScore, while highly correlated with the FICO scores used in our main analysis, may not be identical.²¹ Third, our estimates reflect local average treatment effects for high-leverage, high-risk households and may not generalize to low-risk populations. For households with readily available mortgage credit who may already reside in desirable neighborhoods, an expansion in credit supply might not prompt an immediate or large improvement in neighborhood quality. Fourth, the second stage estimates may partly capture moving-to-opportunity driven by intensive margin effects (i.e. the ability to secure *larger* mortgages), as discussed in Section 4.2.

Finally, we design a placebo test using renters to further alleviate concerns that unobserved differences between low- and high-credit-score individuals, beyond those controlled for using interactive fixed effects, drive the observed shift in neighborhood choice. Specifically, we replicate our analysis for individuals in the credit bureau dataset who remain renters throughout the sample period. As shown in Appendix Table F.3, neighborhood quality for these individuals changes little, and any differences, such as a slight decline in upward mobility, are economically negligible. These findings support the view that the observed improvements in neighborhood quality are unlikely to be explained by unobserved tendencies of low-credit-score individuals to move to better quality locations relative to high-credit-score groups.

²¹Both models predict creditworthiness using similar factors and are highly correlated, but their weighting of inputs may differ, creating small discrepancies. These differences are analyzed [HERE](#).

9 Quantifying Policy Effects on FHA Insurance Premiums

As the FHA policy expands credit volume among high-DTI, low-credit-score (high-risk) borrowers, it reshapes the overall borrower composition and risk profile in this market. What does this mean for the risk borne by the FHA, which acts as an insurer of delinquency risks of this market? How might this change influence FHA's insurance premiums, assuming that the FHA needs to break even on its risk exposure? To quantify these effects, we need to quantify potential credit losses that are driven by the policy change. Specifically, we need to compute the actual and counterfactual delinquency rate per dollar of FHA loan, and the difference between the two.

We write the delinquency rate per dollar of FHA loans as the weighted average of delinquency rates in high- and low-credit-score borrowers after the policy change, with the weight being the value of credit extended in each market segment:

$$D = \frac{delinq_l \times V_l + delinq_h \times V_h}{V_l + V_h}$$

where l and h denote low- and high-credit-score borrowers, respectively, $delinq_l$ ($delinq_h$) is the average delinquency rate of the low-credit-score (high-credit-score) market, and V_l (V_h) as the post-policy credit volume (in dollars) of the low-credit-score (high-credit-score) market.

Similarly, the counterfactual delinquency rate per dollar of FHA loans is

$$\hat{D} = \frac{\hat{delinq}_l \times \hat{V}_l + delinq_h \times V_h}{\hat{V}_l + V_h},$$

where $\hat{delinq}_l$ and $\hat{V}_l$ represents the counterfactual delinquency risk and credit volume in the low-credit-score markets absent of the policy change.

The growth in value-weighted delinquency rate, $g(D)$ can be decomposed into two components: (1) a “Share Effect,” the change in the fraction of low-credit-score borrowers multiplied by the differential delinquency rates between low- and high-credit-score borrowers, and (2) a “Delinquency Effect,” the change in delinquency risk of low-credit-score borrowers due to the shift in their DTI ratio, multiplied by the

counterfactual share of low-credit-score borrowers:

$$g(D) = (D - \hat{D})/\hat{D}$$

$$= \left(\underbrace{\left(\frac{V_l}{V_l + V_h} - \frac{\hat{V}_l}{\hat{V}_l + V_h} \right) (delinq_l - delinq_h)}_{\text{Share Effect}} + \underbrace{(delinq_l - \hat{delinq}_l) \frac{\hat{V}_l}{\hat{V}_l + V_h}}_{\text{Delinquency Effect}} \right) / \hat{D} \quad (11)$$

We gauge $delinq_l$, $delinq_h$, V_l and V_h directly from data. To quantify $\hat{delinq}_l$, we take the weighted average of delinquency rate for each DTI grid d , with the weight being the share of credit amount (in \$) extended to DTI d among low-credit-score borrowers.

$$\hat{delinq}_l = \sum_d delinq_{ld} \frac{\hat{V}_{ld}}{\hat{V}_l}$$

where d represents a DTI integer grid; $delinq_{ld}$ and $\hat{V}_{ld}$ are the average delinquency rates and counterfactual credit volume within credit score category l and DTI category d , respectively.

We compute the counterfactual volumes $\hat{V}_{ld}$ and $\hat{V}_l$ based on the counterfactual distribution of loan counts using the methodology in Section 4.2. To translate loan count into credit volume, we need to estimate loan size m , which we assume is a deterministic function of income and DTI ratios. Within each DTI grid d , we partition borrowers into income deciles, indexed by q , and infer the average loan size m_{ldq} as the average mortgage balance within the DTI-income category $(\{d, q\})$. This gives us $\hat{V}_{ld}$:

$$\hat{V}_{ld} = V_{ld} + \sum_q m_{ldq} (\hat{n}_{ldq}^{post} - n_{ldq}^{post})$$

where m_{ldq} is the average mortgage balance for low-credit-score borrowers within the DTI bin d and income decile q , n_{ldq}^{post} is the number of low-credit-score loans in DTI grid d after the policy, and $\hat{n}_{ldq}^{post}$ is the counterfactual number of loans for low-credit-score borrowers with DTI d . By construction, $\hat{V}_l = \sum_d \hat{V}_{ld}$.

Table 11 reports key quantities from this exercise and walks through the calculation of $g(Risk)$. For variables related to the borrowers affected by the policy, we report their actual and counterfactual values, together with bootstrapped standard errors of the counterfactual estimates. Our estimates suggest that the FHA policy increases the value weighted-delinquency rates among low-credit-score borrowers from 12.8 to 13.2 percentage points, and the share of low-credit-score loans among FHA borrowers from 8.58% to 9.14%. Ultimately, the policy increases the FHA's value-weighted delinquency rate from 5.91% to 5.99%, or an increase of 1.28% from its counterfactual risk level. The Share Effect and Delinquency Effect each

explains roughly half of the overall change.

TABLE 11 ABOUT HERE

Given the above estimates, the FHA’s break-even constraints would require it to increase the mortgage insurance premium (MIP) by the same amount. The mortgage insurance premium (MIP) during our sample period is 175 bps upfront and 85 bps per year for all FHA mortgages regardless of borrower characteristics. For an average loan of 7-year duration, this implies an MIP of 110 bps per year (i.e., $175/7 + 85$). A 1.28% increase of MIP from 110 bps represents an increase of 1.41 bps, which may be absorbed by FHA borrowers through the agency’s premium adjustments or by taxpayers as a transfer. Therefore, our analysis suggests that the credit expansion to low-FICO, high-DTI borrowers likely increases costs for either other FHA borrowers or for taxpayers.

10 Conclusion

Algorithmic underwriting is increasingly important in financial markets today. We study the impacts of the increasing reliance on algorithmic underwriting in a high-risk segment of the U.S. mortgage market by examining an FHA policy that transitioned from pure human underwriting to human-augmented algorithmic underwriting for low-credit-score, high-leverage borrowers. We document that the policy change leads to sizable gains in credit supply without significantly increasing default rates conditional on observables. Our results suggest that a growing reliance on algorithmic underwriting can potentially improve underwriting efficiency, and shed light on *how* algorithms achieve such improvement. In particular, our analysis points to algorithms relieving regulatory concerns for lenders as well as concerns regarding underwriting quality along the financial intermediation chain. Our evidence also suggests that algorithms rely more on standardized data. At the same time, the credit expansion driven by algorithm underwriting is not equally distributed, but is lower among Black and lower-income borrowers. The disparate effects highlight the difficulties associated with increasing financial inclusion for these populations, which could be fruitful directions of future research.

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Table 1. Summary Statistics

Panel A describes the summary statistics of the Ginnie Mae-Endorsements-HMDA matched sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017, excluding August 2016, the month of the policy change. Panel B describes the summary statistics of the sample of individuals in the 1% national representative sample of credit bureau annual records from 2014 to 2019 (excluding 2016) merged with the Opportunity Insights data based on the location of individuals and with the Ginnie Mae data for observations with FHA originations. For Panel B, we also exclude individuals who have ever obtained an FHA mortgage with DTI>50. Variable definitions are provided in [Appendix A](#).

Panel A: Ginnie Mae-HMDA Sample						
	Mean	SD	P25	Median	P75	N
Delinquency	0.059	0.236	0.000	0.000	0.000	705,228
Interest Rate Spread	0.137	0.424	-0.155	0.095	0.390	707,398
FICO	678.376	47.944	644.000	672.000	708.000	707,398
DTI	41.231	9.195	34.960	42.090	48.330	707,398
Low FICO	0.075	0.264	0.000	0.000	0.000	707,398
High DTI	0.460	0.498	0.000	0.000	1.000	707,398
Income	71.646	38.923	45.000	64.000	89.000	705,309
Adjusted Income	1.059	0.570	0.670	0.943	1.311	705,309
Loan Amount	202.532	102.562	130.000	184.000	254.000	707,398
Log(Loan Amount)	12.091	0.512	11.768	12.123	12.441	707,398
Non-Hispanic White	0.609	0.488	0.000	1.000	1.000	707,398
Black	0.119	0.324	0.000	0.000	0.000	707,398
Non-Black	0.881	0.324	1.000	1.000	1.000	707,398

Panel B: Credit Bureau Sample						
	Mean	SD	P25	Median	P75	N
New Purchase FHA	0.001	0.039	0.000	0.000	0.000	10,670,956
Age	51.927	19.274	36.000	51.000	65.000	10,670,956
Female	0.498	0.500	0.000	0.000	1.000	10,670,956
Married	0.562	0.496	0.000	1.000	1.000	10,670,956
Credit Score	684.459	107.349	604.000	692.000	784.000	10,670,956
Up Mob, P25	0.435	0.066	0.391	0.432	0.477	10,653,699
Up Mob, P50	0.505	0.058	0.470	0.506	0.543	10,653,699

Table 2. Quantity of Credit

This table examines the extensive and intensive margin changes in loan origination volume around the changes in underwriting regulations, using the methodology described in Section 4.2. The variables being computed include $\Delta Loans\ Originated$, which refers to the increase in the total number of new purchase loans extended to low-FICO borrowers as a fraction of the counterfactual number of new purchase loans extended to low-FICO borrowers in the absence of the policy; $g(Volume)\ (High\ DTI)$, the increase in the number of new purchase loans to borrowers with FICO below 620 and DTI within 43–50 after the policy change as a fraction of the counterfactual number of new purchase loans for those borrowers in the absence of the policy; $g(Volume)\ (Low\ DTI)$, the increase in the number of new purchase loans extended to borrowers with FICO below 620 and DTI within 35–43 after the policy change as a fraction of the counterfactual number of new purchase loans for those borrowers in the absence of the policy; $\Delta Low\ DTI\ Loans$, the change in low-DTI loans as a fraction of the counterfactual number of new purchase loans for those borrowers in the absence of the policy; and $\Delta Average\ DTI$, the average increase in measured DTI of these new purchase loans as a result of the policy. The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017, excluding August 2016, the month of the policy implementation. DTI is winsorized at the 1st and 99th percentiles and rounded up to the nearest integer. Standard errors are reported in parentheses and are computed from 1,000 bootstrap replications clustered by origination month. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively. Variable definitions are provided in [Appendix A](#).

Threshold:	Baseline	Alternative Specifications		
	$\bar{d} = 35$ (1)	$\bar{d} = 32$ (2)	$\bar{d} = 34$ (3)	$\bar{d} = 36$ (4)
Outcome Variables:				
$\Delta Loans\ Originated$	0.065*** (0.013)	0.068*** (0.018)	0.063*** (0.013)	0.063*** (0.011)
$g(Volume)\ (High\ DTI)$	1.215*** (0.074)	1.192*** (0.083)	1.204*** (0.077)	1.218*** (0.076)
$g(Volume)\ (Low\ DTI)$	-0.161*** (0.019)	-0.131*** (0.021)	-0.152*** (0.019)	-0.176*** (0.020)
$\Delta Low\ DTI\ Loans$	-0.093*** (0.011)	-0.089*** (0.014)	-0.093*** (0.012)	-0.095*** (0.011)
$\Delta Average\ DTI$	1.142*** (0.078)	1.144*** (0.076)	1.142*** (0.078)	1.142*** (0.079)
Number of Observations	707,398	707,398	707,398	707,398

Table 3. Delinquency Rates

This table examines the changes in mortgage delinquency rates around the changes in underwriting regulations. The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017, excluding August 2016, the month of the policy implementation. DTI is winsorized at the 1st and 99th percentiles and rounded up to the nearest integer. Panel A reports results from the DID analysis following Equation 7 for high- and low-DTI loans separately, and Panel B reports the triple-difference analysis following Equation 8 for the combined sample. The outcome variable is *Delinquency*, an indicator equal to one if a loan becomes delinquent for 90 days within two years of issuance. *Treated* is an indicator that equals one if the borrower's credit score is below 620, and zero otherwise. *Post* indicates whether the loan is extended after the regulation change in August 2016. *High DTI* (*Low DTI*) represents a subsample of borrowers with DTI between 43 and 50 (less than or equal to 43). Borrowers with DTI below 35 are unaffected by the policy and are excluded from the sample. Controls include log of loan amount and log of borrower household income. Column (1) of Panel B additionally controls for the remaining triple interaction terms. Standard errors are reported in parentheses and are double clustered by DTI (integer level) and origination month. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively. Variable definitions are provided in [Appendix A](#).

Panel A. Delinquency, Difference-in-Differences						
Subsample:	High DTI ($43 < DTI \leq 50$)			Low DTI ($35 \leq DTI \leq 43$)		
Dep. Var.: <i>Delinquency</i>	(1)	(2)	(3)	(4)	(5)	(6)
<i>Treated</i> $\times$ <i>Post</i>	-0.00996 (0.0121)	-0.0102 (0.0124)	-0.00792 (0.0131)	-0.000640 (0.00572)	-0.000317 (0.00594)	0.00144 (0.00624)
Controls		Yes	Yes		Yes	Yes
Month FE	Yes			Yes		
FICO FE	Yes			Yes		
FICO-DTI FE		Yes	Yes		Yes	Yes
Month-DTI FE		Yes	Yes		Yes	Yes
County FE			Yes			Yes
Lender FE			Yes			Yes
Observations	193092	192570	192226	203345	202698	202375
R^2	0.030	0.032	0.067	0.029	0.032	0.065

Panel B. Delinquency, Triple-Difference			
Dep. Var.: <i>Delinquency</i>	(1)	(2)	(3)
<i>Treated</i> $\times$ <i>High DTI</i> $\times$ <i>Post</i>	0.0165*** (0.00275)	-0.0103 (0.0133)	-0.00907 (0.0132)
<i>High DTI</i> $\times$ <i>Post</i>	-0.000860 (0.00124)	-0.00346 (0.00265)	-0.00189 (0.00395)
<i>Low FICO</i> $\times$ <i>Post</i>	-0.00360 (0.00580)	0.000872 (0.00588)	0.000855 (0.00586)
Controls		Yes	Yes
Month FE	Yes		
FICO FE	Yes		
FICO-DTI FE		Yes	Yes
Month-DTI FE		Yes	Yes
County FE		Yes	Yes
Lender FE			Yes
Observations	396438	395140	395080
R^2	0.029	0.052	0.057

Table 4. Delinquency Rates Effects by Unemployment Rate Change Quartiles

This table examines the changes in mortgage delinquency around the 2016 FHA policy change, across borrowers in regions in each quartile of changes in unemployment rate. Unemployment rate change is the percentage change from year $t - 1$ to t in a county. The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued from August 2015 to August 2017, excluding August 2016, the month of the policy implementation. DTI is winsorized at the 1st and 99th percentiles and rounded up to the nearest integer. The outcome variable is *Delinquency*, an indicator equal to one if a loan becomes delinquent for 90 days within two years of issuance. *Treated* is an indicator that equals one if the borrower's credit score is below 620, and zero otherwise. *Post* indicates whether the loan is extended after the regulation change in August 2016. *High DTI* (*Low DTI*) represents borrowers with DTI between 43 and 50 (35 to 43). Standard errors are reported in parentheses and are double clustered by DTI (integer level) and origination month. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively. Variable definitions are provided in [Appendix A](#).

Panel A: DID, High DTI Loans ($43 < DTI \leq 50$)				
Dep. Var.: <i>Delinquency</i> Sample: Unemp Growth	(1) Qtile 1 (Lowest)	(2) Qtile 2	(3) Qtile 3	(4) Qtile 4 (Highest)
<i>Treated</i> $\times$ <i>Post</i>	-0.0168 (0.0100)	0.00454 (0.0147)	0.00739 (0.0265)	-0.0367 (0.0269)
Controls	Yes	Yes	Yes	Yes
Lender FE	Yes	Yes	Yes	Yes
FICO-DTI FE	Yes	Yes	Yes	Yes
Month-DTI FE	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Observations	48030	48416	48221	46257
R^2	0.097	0.090	0.095	0.098
Panel B: DID, Low DTI Loans ($35 \leq DTI \leq 43$)				
Dep. Var.: <i>Delinquency</i> Sample: Unemp Growth	(1) Qtile 1 (Lowest)	(2) Qtile 2	(3) Qtile 3	(4) Qtile 4 (Highest)
<i>Treated</i> $\times$ <i>Post</i>	-0.00181 (0.0125)	-0.00992 (0.00565)	0.00617 (0.00770)	0.00711 (0.00898)
Controls	Yes	Yes	Yes	Yes
Lender FE	Yes	Yes	Yes	Yes
FICO-DTI FE	Yes	Yes	Yes	Yes
Month-DTI FE	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Observations	49947	50036	49983	51099
R^2	0.093	0.087	0.095	0.091
Panel C: Triple Difference ($DTI \geq 35$)				
Dep. Var.: <i>Delinquency</i> Sample: Unemp Growth	(1) Qtile 1 (Lowest)	(2) Qtile 2	(3) Qtile 3	(4) Qtile 4 (Highest)
<i>Treated</i> $\times$ <i>Post</i> $\times$ <i>High DTI</i>	-0.00807 (0.0176)	0.0184 (0.0186)	0.00238 (0.0266)	-0.0468 (0.0272)
Controls	Yes	Yes	Yes	Yes
Lender FE	Yes	Yes	Yes	Yes
FICO-DTI FE	Yes	Yes	Yes	Yes
Month-DTI FE	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Observations	98487	98918	98635	97920
R^2	0.074	0.072	0.077	0.075

Table 5. Heterogeneity by Race and Income

This table examines the heterogeneity of the credit quantity and delinquency effects of the FHA underwriting policy across racial and income groups. The variables being computed include the changes in credit quantity and delinquency. Credit quantity measures include $\Delta \text{Loans Originated}$, the increase in the number of new purchase loans to low-FICO borrowers as a fraction of the counterfactual number of new purchase loans extended to low-FICO borrowers in the absence of the policy and $g(\text{Volume})$ (*High DTI*), the increase in the total number of new purchase loans extended to borrowers with FICO below 620 and DTI within 43–50 after the policy change as a fraction of the counterfactual number of new purchase loans for low-FICO, high-DTI borrowers in the absence of the policy. The methodology is described in Section 4.2. Delinquency changes are the coefficients on $\text{Treated} \times \text{Post}$ from the delinquency regression, shown in Equation 7, estimated separately from the high-DTI and low-DTI subsamples. *Difference* represents the difference in coefficient estimates between each subsample partition and the statistical significance of the gap. Panel A compares the effects on Black and non-Black borrowers. Panel B compares the effects on high- and low-income borrowers, where “High” (“Low”) represents borrowers with adjusted household income above (below) the sample median. Adjusted household income is computed as borrowers’ own household income scaled by the MSA median. Panel C compares the effects on Black and non-Black borrowers after propensity score weighting by income. The weighting procedure and balance statistics are described in Appendix Section A. The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017, excluding August 2016, the month of the policy implementation. DTI is restricted to between 35 to 50. Standard errors are reported in parentheses and are from 1,000 bootstrap replications clustered by origination month for the bunching analyses, and two-way clustered by origination month and DTI for delinquency analyses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively. Variable definitions are provided in Appendix A.

Panel A: Borrower Race			
Subsample:	Black	Non-Black	Difference (Black - Non-Black)
Outcome: <i>Volume Change</i> <i>ΔLoans Originated</i>	-0.010 (0.038)	0.073*** (0.013)	-0.083** (0.039)
<i>g(Volume) (High DTI)</i>	0.859*** (0.151)	1.263*** (0.077)	-0.404** (0.160)
Outcome: <i>Delinquency Change (Coeff on Treated × Post)</i>			
High DTI:	0.026 (0.029)	-0.016 (0.011)	0.042 (0.026)
Low DTI:	-0.000 (0.012)	0.001 (0.006)	-0.002 (0.013)
Panel B: Borrower Income			
Subsample:	High Income	Low Income	Difference (High - Low)
Outcome: <i>Volume Change</i> <i>ΔLoans Originated</i>	0.067*** (0.017)	0.045** (0.023)	0.022 (0.030)
<i>g(Volume) (High DTI)</i>	1.595*** (0.129)	0.778*** (0.077)	0.817*** (0.161)
Outcome: <i>Delinquency Change (Coeff on Treated × Post)</i>			
High DTI:	-0.015 (0.016)	0.003 (0.012)	-0.018 (0.011)
Low DTI:	0.004 (0.008)	0.002 (0.008)	0.002 (0.012)
Panel C: Race Partition After Propensity Score Weighting by Income			
Subsample	Black	Non-Black	Difference (Black - Non-Black)
Outcome: <i>Volume Change</i> <i>ΔLoans Originated</i>	-0.011 (0.038)	0.072*** (0.012)	-0.083** (0.038)
<i>g(Volume) (High DTI)</i>	0.855*** (0.151)	1.227*** (0.072)	-0.372** (0.161)
Outcome: <i>Delinquency Change (Coeff on Treated × Post)</i>			
High DTI:	0.026 (0.029)	-0.015 (0.012)	0.041 (0.027)
Low DTI:	-0.000 (0.012)	0.002 (0.006)	-0.002 (0.013)

Table 6. Mechanism: Regulatory Concerns

This table examines the heterogeneity effects of the FHA underwriting policy across lenders facing high and low regulatory concerns. $\Delta\text{Loans Originated}$ refers to the increase in the total number of new purchase loans extended to low-FICO borrowers as a fraction of the counterfactual number of new purchase loans extended to low-FICO borrowers in the absence of the policy. $g(\text{Volume})$ (*High DTI*) refers to the increase in the total number of new purchase loans extended to borrowers with FICO below 620 and DTI within 43–50 after the policy change as a fraction of the counterfactual number of new purchase loans for low-FICO, high-DTI borrowers in the absence of the policy. The methodology is described in Section 4.2. *Difference* represents the difference in coefficient estimates between each subsample partition and the statistical significance of the gap. Panel A reports results for bank and non-bank originators; Panel B reports results for bank and non-bank issuers; and Panel C reports results among loans produced by bank originators or issuers that have and have not faced litigation related to their FHA mortgage practices. The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017, excluding August 2016, the month of the policy implementation. DTI is restricted to between 35 to 50. Standard errors are reported in parentheses and are from 1,000 bootstrap replications clustered by origination month. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively. Variable definitions are provided in [Appendix A](#).

Panel A: Bank vs Non-Bank Originators			
Subsample:	Bank Originator	Non-Bank Originator	Difference (Bank – NonBank)
Outcome: <i>Volume Change</i>			
$\Delta\text{Loans Originated}$	0.132*** (0.036)	0.054*** (0.012)	0.079** (0.034)
$g(\text{Volume})$ (<i>High DTI</i>)	1.856*** (0.253)	1.160*** (0.079)	0.696*** (0.239)
Panel B: Bank vs Non-Bank Issuers			
Subsample:	Bank Issuer	Non-Bank Issuer	Difference (Bank – NonBank)
Outcome: <i>Volume Change</i>			
$\Delta\text{Loans Originated}$	0.153*** (0.053)	0.052*** (0.015)	0.101* (0.057)
$g(\text{Volume})$ (<i>High DTI</i>)	1.788*** (0.271)	1.142*** (0.088)	0.646** (0.299)
Panel C: False Claims Act Litigated vs. Non-Litigated Banks			
Subsample:	Litigated	Non-Litigated	Difference (Litigated – Non)
Outcome: <i>Volume Change</i>			
$\Delta\text{Loans Originated}$	0.212*** (0.062)	0.092** (0.036)	0.120* (0.068)
$g(\text{Volume})$ (<i>High DTI</i>)	2.004*** (0.355)	1.611*** (0.249)	0.393 (0.396)

Table 7. Mechanism: Information Frictions in Financial Intermediation

This table examines the heterogeneity effects of the FHA underwriting policy across lenders that are subject to high and low information frictions in the financial intermediation chain. $\Delta \text{Loans Originated}$ refers to the increase in the total number of new purchase loans extended to low-FICO borrowers as a fraction of the counterfactual number of new purchase loans extended to low-FICO borrowers in the absence of the policy. $g(\text{Volume}) (\text{High DTI})$ refers to the increase in the total number of new purchase loans extended to borrowers with FICO below 620 and DTI within 43–50 after the policy change as a fraction of the counterfactual number of new purchase loans for low-FICO, high-DTI borrowers in the absence of the policy. The methodology is described in Section 4.2. *Difference* represents the difference in coefficient estimates between each subsample partition and the statistical significance of the gap. Panel A reports results for aggregators-dependent originators (“indirect sellers”) and all other originators (“other originators”). Panel B repeats the analysis, after using the covariance-balanced propensity score (CBPS) weighting approach to balance indirect sellers’ lender type (i.e., bank vs non-bank) and bank size (i.e., asset size) to match those of direct sellers. This procedure is described in Appendix Section D. The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017, excluding August 2016, the month of the policy implementation. DTI is restricted to between 35 to 50. Standard errors are reported in parentheses and are from 1,000 bootstrap replications clustered by origination month. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively. Variable definitions are provided in Appendix A.

Panel A: Indirect Sellers vs Other Originators			
Subsample:	Indirect Sellers	Other Originators	Difference (Indirect Sellers – Other Originators)
Outcome: <i>Volume Change</i> <i>$\Delta \text{Loans Originated}$</i>	0.115*** (0.030)	0.042*** (0.014)	0.073** (0.032)
<i>g(Volume) (High DTI)</i>	1.558*** (0.149)	1.016*** (0.085)	0.542*** (0.171)
Panel B: Indirect Sellers vs Other Originators, Weighted by Propensity Scores			
Subsample:	Indirect Sellers	Other Originators	Difference (Indirect Sellers – Other Originators)
Outcome: <i>Volume Change</i> <i>$\Delta \text{Loans Originated}$</i>	0.115*** (0.030)	0.034 (0.026)	0.081* (0.042)
<i>g(Volume) (High DTI)</i>	1.558*** (0.149)	1.059*** (0.122)	0.499** (0.215)

Table 8. Mechanism: Availability of Standardized Data

This table examines the heterogeneity effects of the FHA underwriting policy across borrowers with differential availability of standardized data. Specifically, we compare first time homebuyers and repeat homebuyers, sole applicants and those applying with a co-applicant, and borrowers from high (above median) and low self-employment share regions. $\Delta \text{Loans Originated}$ refers to the increase in the total number of new purchase loans extended to low-FICO borrowers as a fraction of the counterfactual number of new purchase loans extended to low-FICO borrowers in the absence of the policy. $g(\text{Volume})$ (*High DTI*) refers to the increase in the total number of new purchase loans extended to borrowers with FICO below 620 and DTI within 43–50 after the policy change as a fraction of the counterfactual number of new purchase loans for low-FICO, high-DTI borrowers in the absence of the policy. The methodology is described in Section 4.2. *First Time Homebuyer* and *Repeat Homebuyer* indicate samples of loans where the borrower is labeled as the first time homebuyer or not. *Co-Applicant* and *Sole Applicant* represent samples of loans where borrowers have and do not have a co-applicant, respectively. *High (Low) Self-Employment* represent samples where the fraction of income in a zip code arising from self-employment is above (below) the sample median, respectively. The fraction of self-employment income comes from the IRS SOI data. *Difference* represents the difference in coefficient estimates between each subsample partition and the statistical significance of the gap. For each outcome variable, we report both the raw estimates from bunching in the “Unweighted” columns and the estimates obtained after conducting propensity-score weighting on borrower race and income in the “Weighted” columns. The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017, excluding August 2016, the month of the policy implementation. DTI is restricted to between 35 to 50. Standard errors are reported in parentheses and are from 1,000 bootstrap replications clustered by origination month. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively. Variable definitions are provided in [Appendix A](#).

Dep. Var.:	$\Delta \text{Loans Originated}$		$g(\text{Volume})$ (<i>High DTI</i>)	
Estimation Method:	Unweighted	Weighted	Unweighted	Weighted
First Time Homebuyer	0.038*** (0.014)	0.038*** (0.013)	1.053*** (0.088)	1.046*** (0.071)
Repeat Homebuyer	0.155*** (0.033)	0.171*** (0.038)	1.659*** (0.185)	1.586*** (0.221)
<i>Difference</i>	-0.117*** (0.036)	-0.133*** (0.041)	-0.605*** (0.220)	-0.540*** (0.221)
Sole Applicant	0.047*** (0.001)	0.046*** (0.016)	0.859*** (0.000)	0.856*** (0.071)
Co-Applicant	0.090** (0.035)	0.086*** (0.034)	1.798*** (0.093)	1.471*** (0.167)
<i>Difference</i>	-0.043 (0.035)	-0.040 (0.038)	-0.939*** (0.003)	-0.616*** (0.194)
High Self-Employment	0.019 (0.021)	0.018 (0.018)	1.040*** (0.093)	1.036*** (0.097)
Low Self-Employment	0.094*** (0.016)	0.105*** (0.016)	1.298*** (0.108)	1.456*** (0.129)
<i>Difference</i>	-0.075** (0.026)	-0.087*** (0.026)	-0.258*** (0.135)	-0.419*** (0.151)

Table 9. Interest Rate Spreads

This table examines changes in interest rate spreads around the change in the FHA underwriting regulations. The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017. DTI is winsorized at the 1st and 99th percentiles and rounded up to the nearest integer. Panel A reports results from the DID analysis following Equation 7 for subsamples of high- and low-DTI loans separately, and Panel B reports the triple-difference analysis following Equation 8 using the combined sample. The dependent variable is the interest rate spreads on a mortgage relative to the Freddie Mac Survey rate. *Treated* is an indicator that equals one if the borrower's credit score is below 620, and zero otherwise. *Post* indicates whether the loan is extended after the regulation change in August 2016. *High DTI* (*Low DTI*) represents a subsample of borrowers with DTI above 43 (less than or equal to 43). Borrowers with DTI below 35 are unaffected by the policy and are excluded from the sample. Controls include log of loan amount and log of borrower household income. Standard errors are reported in parentheses and are double clustered by DTI (integer level) and origination month. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively. Variable definitions are provided in [Appendix A](#).

Panel A. Interest Rate Spreads, Difference-in-Differences

Subsample:	High DTI ($43 < DTI \leq 50$)			Low DTI ($35 \leq DTI \leq 43$)		
Dep. Var.: <i>Interest Rate Spreads</i>	(1)	(2)	(3)	(4)	(5)	(6)
<i>Treated</i> $\times$ <i>Post</i>	0.00155 (0.0234)	0.00523 (0.0236)	0.00399 (0.0220)	0.0394*** (0.0105)	0.0336** (0.0109)	0.0225 (0.0120)
Controls		Yes	Yes		Yes	Yes
Month FE	Yes			Yes		
FICO-DTI FE		Yes	Yes		Yes	Yes
FICO FE	Yes			Yes		
Month-DTI FE		Yes	Yes		Yes	Yes
County FE			Yes			Yes
Lender FE			Yes			Yes
Observations	193723	193195	192848	204076	203415	203092
R^2	0.230	0.245	0.472	0.255	0.272	0.502

Panel B. Interest Rate Spreads, Triple-Difference

Dep. Var.: <i>Interest Rate Spreads</i>	(1)	(2)	(3)
<i>Treated</i> $\times$ <i>High DTI</i> $\times$ <i>Post</i>	0.0183 (0.0111)	-0.0277* (0.0139)	-0.0194 (0.0174)
<i>High DTI</i> $\times$ <i>Post</i>	-0.00462 (0.00729)	-0.0309*** (0.00695)	0.0298*** (0.00411)
<i>Treated</i> $\times$ <i>Post</i>	0.0332** (0.0115)	0.0343** (0.0144)	0.0233* (0.0127)
Controls		Yes	Yes
Month FE	Yes		
FICO-DTI FE		Yes	Yes
FICO FE	Yes		
Month-DTI FE		Yes	Yes
County FE			Yes
Lender FE			Yes
Observations	397801	396612	396421
R^2	0.245	0.261	0.484

Table 10. Mortgage Access and Neighborhood Choices: Moving to Opportunity

This table reports the results of the 2SLS approach, which examines the effect of mortgage access on “moving to opportunity.” The sample includes individuals in the 1% national representative sample of credit bureau annual records from 2014 to 2019 (excluding 2016), and is merged with the upward mobility and neighborhood characteristics measures obtained from [Chetty et al. \(2025\)](#) based on the location of individuals, and with the Ginnie Mae data for observations with FHA originations. The unit of observation is an individual-year. We exclude individuals who have ever obtained an FHA mortgage with DTI>50 and weight observations by the inverse of the annual match probabilities as described in Section 8.2. Panel A reports results from the first-stage regressions, where the dependent variable is an indicator for whether an individual obtains a new purchase FHA mortgage in a year. Panel B reports the second-stage results, where the dependent variable is year-on-year changes in the upward mobility of an individual’s census tract of residence. *Up Mob, P25* (*Up Mob, P50*) measures the household income rank for children of parents at the 25th (50th) percentile. *Treated* (2015) is an indicator that equals one if the borrower’s Vantage Score is below 620 in 2015, and zero otherwise. *Post* indicates whether the loan is extended after the regulation change in 2016. Individual characteristics include indicators for gender, marital status, and *Treated* (2015). Age groups are: 18–35, 35–45, and above 45. A placebo test of this analysis for households who are renters throughout the sample is shown in Appendix Table F.3. Standard errors, clustered by county, are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively. Variable definitions are provided in [Appendix A](#).

Panel A. First Stage, Obtaining FHA Mortgage

Dep. Var.: <i>New Purchase FHA</i>	(1)	(2)
<i>Treated</i> (2015) $\times$ <i>Post</i>	0.0017*** (0.0001)	0.0019*** (0.0001)
Individual Char	Yes	Yes
Credit Score FE	Yes	Yes
Age Group FE	Yes	
Zip-Year FE	Yes	Yes
Age Group-Year FE		Yes
Married-Year FE		Yes
Gender-Year FE		Yes
Observations	10,670,956	10,670,956
R ²	0.02	0.02
F-statistic	164.71	164.44

Panel B. Second Stage, Moving to Opportunity

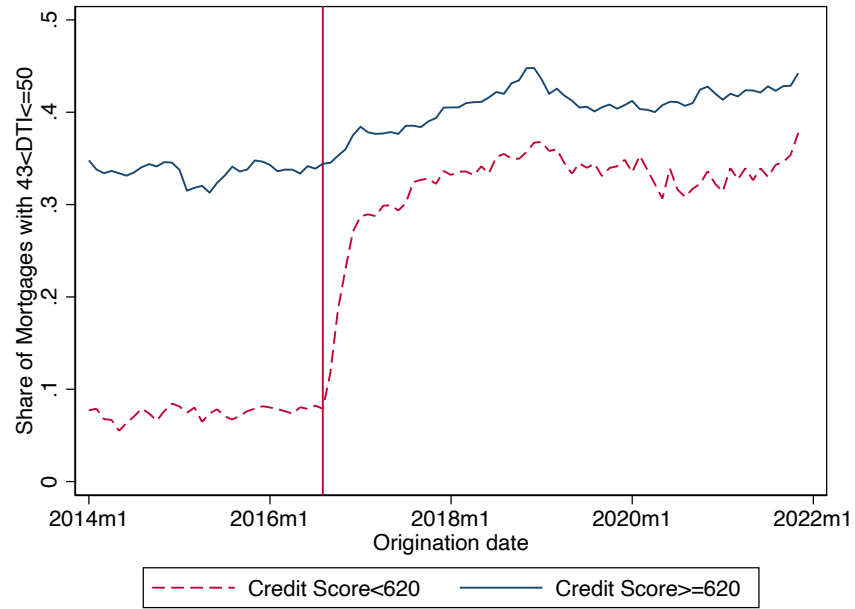
Dep. Var:	<i>d</i> (<i>Up Mob, P25</i>)		<i>d</i> (<i>Up Mob, P50</i>)	
	(1)	(2)	(3)	(4)
<i>New Purchase FHA</i>	0.175*** (0.028)	0.133*** (0.025)	0.175*** (0.027)	0.135*** (0.023)
Individual Char	Yes	Yes	Yes	Yes
Credit Score FE	Yes	Yes	Yes	Yes
Age Group FE	Yes		Yes	
Zip-Year FE	Yes	Yes	Yes	Yes
Age Group-Year FE		Yes		Yes
Married-Year FE		Yes		Yes
Gender-Year FE		Yes		Yes
Observations	10,642,872	10,642,872	10,642,872	10,642,872

Table 11. Effect of Policy on the FHA's Value-Weighted Delinquency Rate

This table examines the changes in the FHA's value-weighted delinquency rate due to the policy. The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued over the treatment period of September 2016 through August 2017 with DTI less than or equal to 50. Delinquency rates are measured as 90-day, 2-year delinquency rates. Standard errors are reported in parentheses and are from 1,000 bootstrap replications clustered by origination month. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

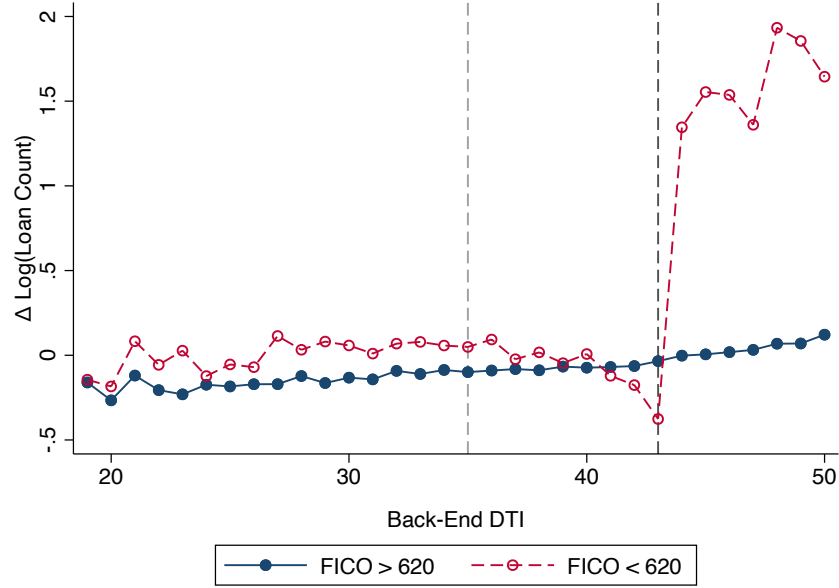
Variables	Actual	Counterfactual
Low-Credit Score Delinquency Rates (delinq_l and $\hat{\text{delinq}}_l$)	0.132	0.128 (0.00107)
Share of Low-Credit-Score Borrowers ($\frac{V_l}{V_l+V_h}$ and $\frac{\hat{V}_l}{\hat{V}_l+V_h}$)	0.0914	0.0858 (0.00176)
High-Credit Score Delinquency Rates (delinq_h)	0.0526	
Share Effect ($\frac{V_l}{V_l+V_h} - \frac{\hat{V}_l}{\hat{V}_l+V_h}$) $\times$ ($\text{delinq}_l - \text{delinq}_h$)	0.000440** (0.000140)	
Delinquency Effect ($\text{delinq}_l - \hat{\text{delinq}}_l$) $\times$ ($\frac{\hat{V}_l}{\hat{V}_l+V_h}$)	0.000326*** (0.0000913)	
Combined Effects	0.000766*** (0.000168)	
Growth in Value-Weighted Delinquency Rate ($g(D) = \text{Combined Effects}/\hat{D}$)	0.0128*** (0.00281)	

Figure 1. Effect of the policy change on the share of high DTI mortgages ($43 < \text{DTI} \leq 50$)



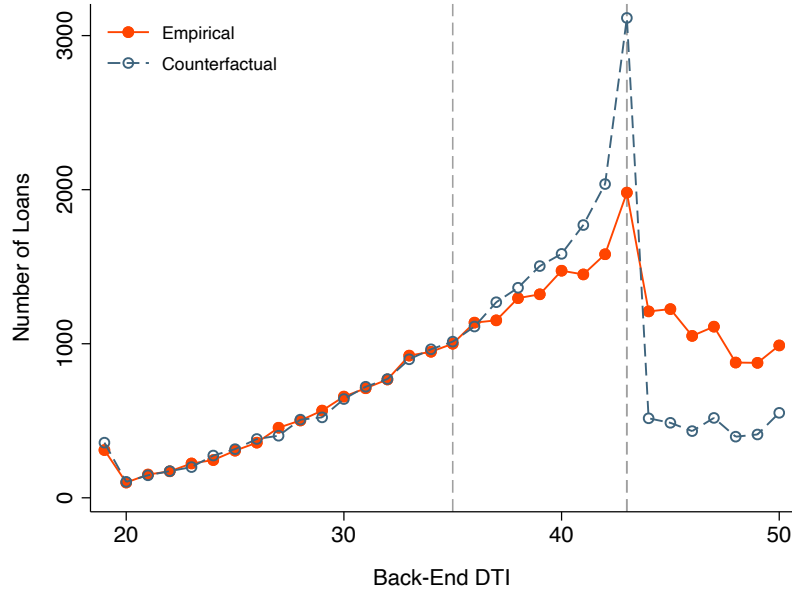
Note: This figure plots the share of FHA new purchase mortgages with DTI greater than 43 and less than or equal to 50 by their month of origination. The sample is the full sample of FHA loans in our Ginnie Mae data from January 2014 to January 2022 with DTI less than or equal to 50. The red, dashed line represents borrowers with credit score less than 620 (i.e., low-credit-score) and the blue solid line represents borrowers with credit score greater than or equal to 620 (i.e., high-credit-score). The policy month of August 2016 is marked via a vertical red line. The effect of the policy change in our Ginnie Mae-Endorsements-HMDA sample is shown in Appendix Figure B.1.

Figure 2. Loan growths around the FHA removal of human underwriting mandate



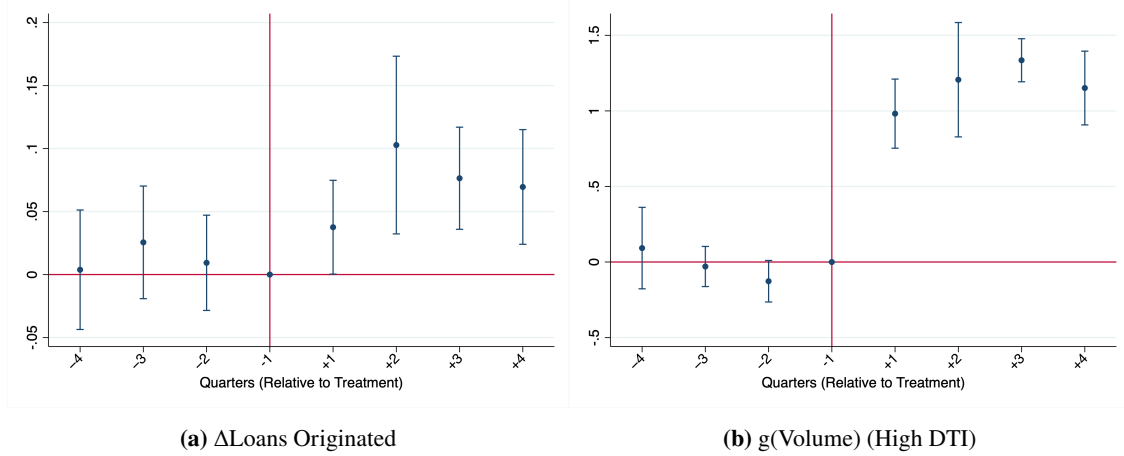
Note: This figure plots the log difference of the number of loans between the 12 months after the policy and the number of loans 12 months before the policy by DTI. The loan counts include all FHA single-family, non-manufactured housing new purchase mortgages in our Ginnie Mae-Endorsements-HMDA sample. DTI is winsorized at the 1st and 99th percentiles and rounded up to the nearest integer. Dashed lines are drawn at DTI equals 43, above which the policy takes into effect, and at DTI equals 35, at or below which we assume is unaffected by the policy for our baseline bunching analysis. We show that this assumption along with a parallel trends assumption fits the data well for $DTI \leq 35$ borrowers in Figure 3.

Figure 3. Effect of the policy change on loan quantities by DTI



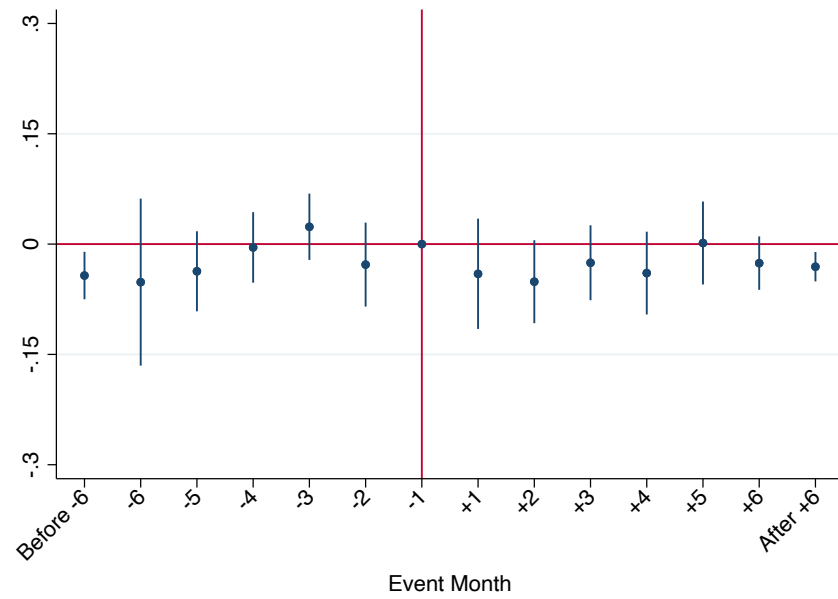
Note: This figure plots empirical and counterfactual numbers of FHA single-family, non-manufactured housing new purchase mortgages in our Ginnie Mae-Endorsements-HMDA sample during 12 months after the policy based on the methodology described in Section 4.2. DTI is winsorized at the 1st and 99th percentiles and rounded up to the nearest integer. Dashed lines are drawn at DTI equals 43, above which the policy takes into effect, and at DTI equals 35, at or below which we assume is unaffected by the policy for our baseline bunching analysis. We show in this figure that this assumption along with a parallel trends assumption fits the data well for $DTI \leq 35$. We plot the empirical and counterfactual numbers for placebo periods surrounding our policy in Appendix Figure B.3, and the raw data for high-FICO and low-FICO borrowers in Appendix Figure B.2.

Figure 4. Dynamic effects of credit expansion



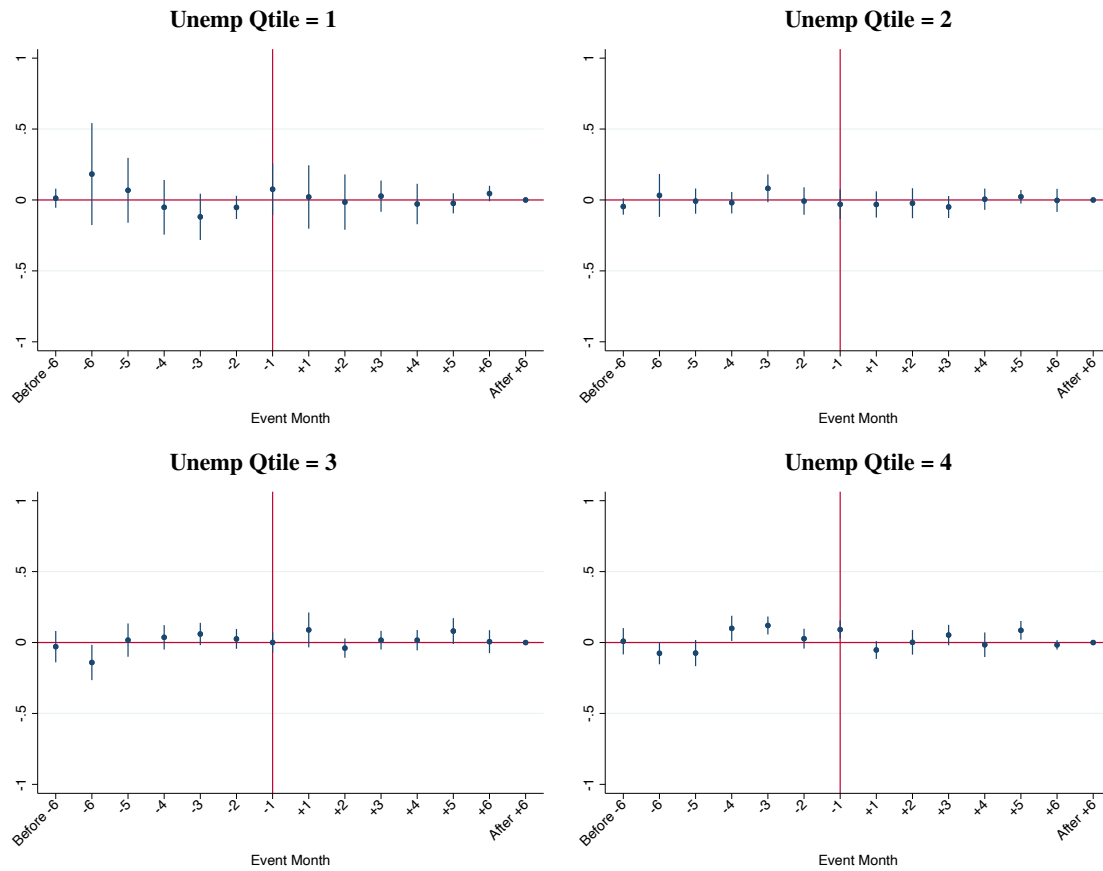
Note: We estimate the dynamic effects implied by the empirical and counterfactual distributions produced from the bunching analyses based on the methodology described in Section 4.2, with the quarter prior to treatment (i.e. April 2016 – June 2016) used as the base period for the estimation. In Panel (a), we plot the quarterly estimates relative to the base period analogous to the Δ Loans Originated number reported in Table 2 as the coefficient estimates. In Panel (b), we plot the quarterly estimates relative to the base period analogous to the $g(\text{Volume})$ (High DTI) number reported in Table 2 as the coefficient estimates. We also plot two-tailed 95% confidence intervals based on standard errors from 1,000 bootstraps clustered by origination month.

Figure 5. Trends in delinquency rates



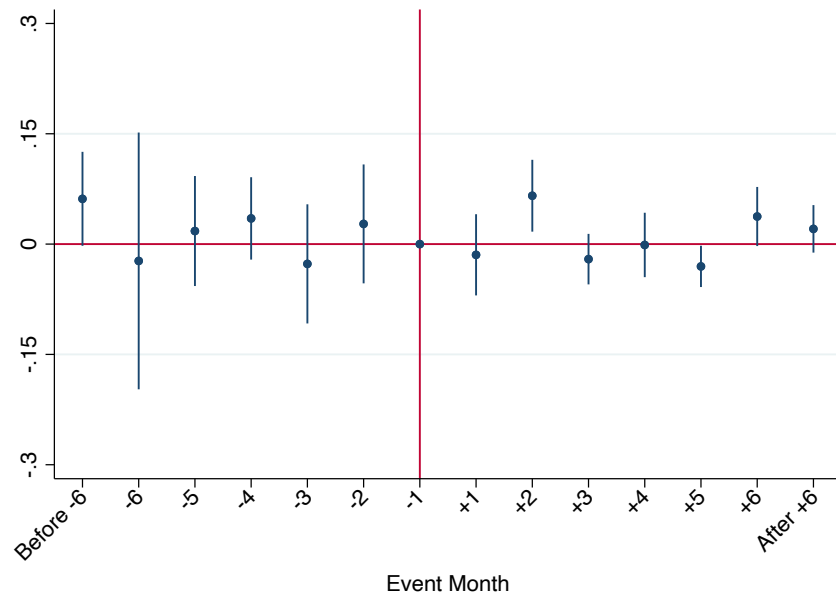
Note: We estimate dynamic triple-difference regressions and plot the coefficient estimates on the event month indicators and the two-tailed 95% confidence intervals. The outcome variable is the 90-day delinquency indicator measured in the two years post origination. The fixed effects and control variables are the same as those used in Table 3 Panel B Column (3). We use the month prior to August 2016 as the base period for estimation (Event Month = -1).

Figure 6. Trends in delinquency by quartiles of unemployment rate change



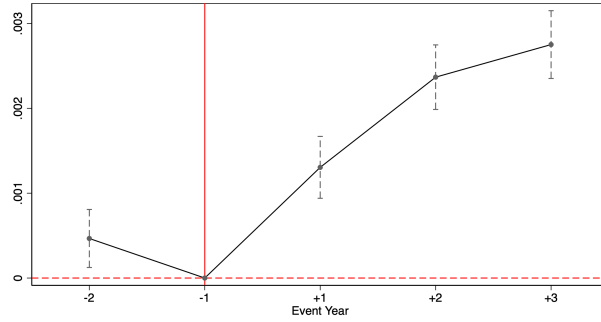
Note: We estimate dynamic triple-difference regressions and plot the coefficient estimates on the event month indicators and the two-tailed 95% confidence intervals. The outcome variable is 90-day delinquency indicator measured in the two years post origination. The fixed effects and control variables used are the same as those used in column (3), Panel B of Table 3. We use the month prior to August 2016 as the base period for estimation (Event Month = -1). We split the samples based on the quartile of unemployment rate growth, measured as the year-on-year change in unemployment rate in the county of the loan application.

Figure 7. Trends in interest rate spreads

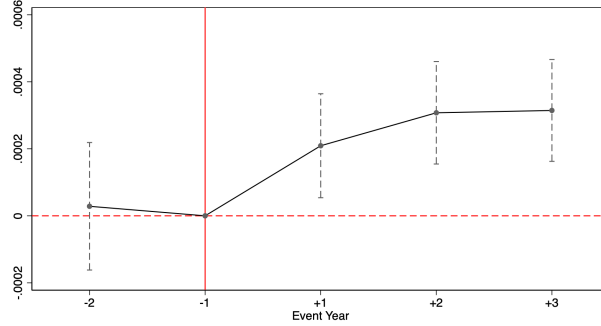


Note: We estimate dynamic triple-difference regressions and plot the coefficient estimates on the event month indicators and the two-tailed 95% confidence intervals. The outcome variable is mortgage interest rate spread. The fixed effects and control variables are the same as those used in Table 9 Panel B Column (3). We use the month prior to August 2016 as the base period for estimation (Event Month = -1).

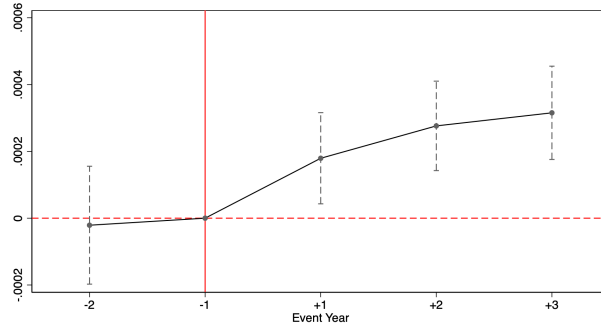
Figure 8. Trends in neighborhood quality



Panel A. New Purchase FHA



Panel B. d(Up Mob, P25)



Panel C. d(Up Mob, P50)

Note: We estimate the dynamic effect of the FHA policy on new FHA purchases and neighborhood quality using credit bureau data. The dots represent coefficient estimates of *Treated* interacted with indicators of event years, ranging from year 2014 (two years before) to 2019 (three years after). The vertical intervals represent two-tailed 95% confidence intervals. The dependent variables are listed at the top of each graph, and described in the notes to Table 10. The specification employs the same sample, fixed effects, control variables, and standard error clustering structure as the most saturated specifications in Table 10. The base year for the estimations is the year prior to the policy implementation.

Appendix A Variable Definitions

- *Delinquency*: An indicator variable for whether a loan is ever 90 days or more than 90 days delinquent within two years of the first payment date.
- *FICO*: The integer FICO score of the borrower.
- *Low FICO*: An indicator for whether a borrower's FICO score is below 620.
- *DTI*: a borrower's debt-to-income ratio, rounded up to the nearest integer.
- *High DTI*: An indicator for whether the borrower's DTI is greater than 43.
- *Δ Loans Originated*: The increase in the total number of new purchase loans extended to borrowers with below-620 FICO scores as a fraction of the counterfactual number of new purchase loans for borrowers with below-620 FICO in the absence of the policy. The estimation methodology is described in Section 4.2
- *$g(\text{Volume})$ (High DTI)*: the increase in the number of new purchase loans to borrowers with FICO below 620 and DTI within 43–50 after the policy change as a fraction of the counterfactual number of new purchase loans for borrowers with FICO below 620 and DTI within 43–50 in the absence of the policy. The estimation methodology is described in Section 4.2.
- *$g(\text{Volume})$ (Low DTI)*: the increase in the number of new purchase loans extended to borrowers with FICO below 620 and DTI within 35–43 after the policy change as a fraction of the counterfactual number of new purchase loans for borrowers with FICO below 620 and DTI within 35–43 in the absence of the policy. The estimation methodology is described in Section 4.2.
- *Δ Low DTI Loans*: the change in low-DTI loans as a fraction of the counterfactual number of new purchase loans for borrowers with FICO below 620 in the absence of the policy.
- *Δ Average DTI*: the average increase in measured DTI of new purchase loans among borrowers with FICO below 620 relative to the counterfactual level of average DTI ratio absent of the policy.
- *Income*: A borrower's application income in thousands.
- *Adjusted Income*: A borrower's application income scaled by the median income of the MSA where the borrower is located.
- *Loan Amount*: The amount of the loan in thousands.
- *Black*: An indicator variable that takes the value of one if the borrower's race is reported as Black.
- *Bank (Non-Bank)*: Banks are HMDA lenders that can be linked to an entity in the commercial bank Call Report through Robert Avery's crosswalk: <https://sites.google.com/site/neilbhutta/data>. Other lenders are classified as non-banks.

- *Litigated Banks*: Bank lenders that have settled their FHA-fraud litigations before the pre-policy window, based on the list from [Frame et al. \(Forthcoming\)](#).
- *Indirect Sellers*: Lenders that sell more than 95% of their originated FHA mortgages, excluding those held in their portfolios, to entities other than Ginnie Mae or affiliated institutions at the time of HMDA reporting.
- *First Time Homebuyer*: Loans extended to borrowers who are first time homeowners.
- *Sole Applicant*: Loans issued to borrowers without a co-applicant.
- *Co-Applicant*: Loans issued to borrowers with a co-applicant.
- *High Self-Employment*: An indicator for whether the fraction of income in a zip code arising from self-employment exceeds the sample median. The fraction of self-employment income in a zip code comes from the IRS SOI data.
- *Interest Rate Spread*: The mortgage interest rate spread over the 30-year Freddie Mac survey rate.
- *New Purchase FHA*: An indicator variable that takes the value of one if an individual has obtained a new FHA mortgage purchase in a given year.
- *Age*: The age of the individual.
- *Female*: An indicator variable that takes the value of one if an individual is a female.
- *Married*: An indicator variable that takes the value of one if an individual is married.
- *Credit Score*: The credit score of the borrower reported in the credit bureau data.
- *Up Mob, P25 (Up Mob, P50)*: the household income rank for children of parents at the 25th (50th) percentile in a zip code from [Chetty et al. \(2025\)](#).

Internet Appendix for “Algorithmic Underwriting in High Risk Mortgage Markets”

This appendix supplements the empirical analysis of this paper. Below is a list of the sections contained in this appendix.

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A Sample Construction and Choices

A The Ginnie Mae-HMDA match

We merge the Ginnie Mae and HMDA data using FHA endorsements as an intermediate link. The FHA endorsements data contains the universe of single-family mortgages insured by the FHA and is published on the U.S. Department of Housing and Urban Development (HUD)’s website.¹

To merge the Ginnie Mae data and FHA endorsements, we take a two-step approach. In the first step, we exact match on the property state, interest rate, the balance of the mortgage rounded down to the nearest 1000, whether the mortgage is fixed rate, the mortgage purpose, and whether the mortgage’s endorsement month is within 3 months of origination. In the second step, we take the unique matches from the first step and identify a seller-lender correspondence by keeping only the Ginnie Mae sellers that are among the top 10 sellers associated with the matched endorsement FHA lender (sponsor) and that have a market share of at least 5% associated with the matched endorsement FHA lender (sponsor). As the average seller market share is 57% for the top seller associated with each sponsor, this is a fairly permissive restriction. Overall, we were able to uniquely merge 62% of Ginnie Mae loans to FHA endorsements.

To merge the HMDA data and FHA endorsements, we also take a two-step approach. In the first step, we match on whether the property’s zip code in the endorsement data contains a Census tract with a positive residential ratio that is associated with the HMDA data as found in HUD’s March 2016 crosswalk,² the balance of the mortgage rounded to the nearest 1000, the mortgage purpose, and whether the mortgage’s endorsement month is either in the HMDA’s year of origination or within 3 months of it. In the second step, we take the unique matches from the first step and identify a lender-FHA sponsor correspondence by keeping only the HMDA lenders that have a market share of at least 20% associated with the matched endorsement FHA sponsor. As in theory the correspondence between HMDA lenders and FHA sponsors should be one-to-one and the average market share for the top lender associated with each sponsor in our first step matched sample is 91%, this is a fairly permissive restriction. Overall, we were able to uniquely merge 81% of FHA endorsements to HMDA loans.

Linking the datasets together, we obtain a total unique match rate of 49%. Restricting to new purchase, single-family, non-manufactured housing mortgages in our sample period, the match rate is 43%. We use only the uniquely matched loans for our empirical analyses. To alleviate concerns about match quality, we also run our extensive margin and loan performance analysis on the Ginnie Mae sample alone, and obtain similar qualitative results. These results are tabulated in Table A.1.

¹https://www.hud.gov/program_offices/housing/rmra/oe/rpts/sfsnap/sfsnap.

²https://www.huduser.gov/portal/datasets/usps_crosswalk.html

Table A.1. Results from the Ginnie Mae Sample

This table examines the changes in loan origination volume and loan performance around the changes in FHA underwriting regulations. The sample is our Ginnie Mae sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017, excluding August 2016, the month of the policy implementation. DTI is winsorized at the 1st and 99th percentiles and rounded up to the nearest integer. Panel A replicates Table 2, based on the methodology described in Section 4.2. Panel B examines the changes in mortgage delinquency rates around the FHA policy change following Equation 7. The analysis follows the one in Table 3, with the exception that we do not control for borrower income and county and lender fixed effects because those variables come from HMDA data. Instead, we include state fixed effects in Columns (3) and (6). Standard errors in Panel A are reported in parentheses and are computed from 1,000 bootstrap replications clustered by origination month. Standard errors in Panel B are double clustered by DTI (integer level) and origination month. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Panel A. Effects on Credit Quantity				
	Baseline	Alternative Specifications		
Threshold:	$\bar{d} = 35$	$\bar{d} = 32$	$\bar{d} = 34$	$\bar{d} = 36$
$\Delta \text{Loans Originated}$	0.065*** (0.011)	0.061*** (0.013)	0.064*** (0.011)	0.066*** (0.011)
$g(\text{Volume}) (\text{High DTI})$	1.031*** (0.049)	0.993*** (0.051)	1.015*** (0.048)	1.037*** (0.052)
$g(\text{Volume}) (\text{Low DTI})$	-0.117*** (0.020)	-0.099*** (0.019)	-0.109*** (0.018)	-0.124*** (0.020)
$\Delta \text{Low DTI Loans}$	-0.065*** (0.011)	-0.066*** (0.012)	-0.065*** (0.011)	-0.064*** (0.011)
$\Delta \text{Average DTI}$	1.180*** (0.073)	1.171*** (0.071)	1.174*** (0.072)	1.178*** (0.075)
Number of Observations	1,692,934	1,692,934	1,692,934	1,692,934

Panel B. Effects on Delinquency						
Sample	High DTI ($43 < DTI \leq 50$)			Low DTI ($35 \leq DTI \leq 43$)		
Dep. Var.: <i>Delinquency</i>	(1)	(2)	(3)	(4)	(5)	(6)
<i>Treated</i> $\times$ Post	-0.00250 (0.0108)	-0.00334 (0.0110)	-0.00120 (0.0106)	0.00335 (0.00386)	0.00455 (0.00391)	0.00543 (0.00384)
Controls		Yes	Yes		Yes	Yes
Month FE	Yes			Yes		
FICO FE	Yes			Yes		
FICO-DTI FE		Yes	Yes		Yes	Yes
Month-DTI FE		Yes	Yes		Yes	Yes
State FE			Yes			Yes
Observations	465363	465352	465351	487894	487885	487885
R^2	0.028	0.029	0.040	0.028	0.029	0.038

B Algorithmic underwriting in the DTI>50 region

In this subsection we examine the risk management effectiveness of algorithms in the DTI>50 region for low-FICO borrowers. We note that human underwriting is not allowed for DTI>50, low-FICO borrowers throughout our sample period. On the other hand, algorithms are allowed to underwrite DTI>50, low-FICO

borrowers post-policy. Therefore, the borrowers in this DTI region allows us to analyze the risk management effectiveness of the FHA’s underwriting algorithm when they alone can approve borrowers.

We begin by adapting the [DeFusco et al. \(2020\)](#) methodology to quantify the credit expansion in this region, and then examine delinquency rates. In theory, there should be no loans in low FICO, DTI>50 region pre-policy, as humans were unable to underwrite in this region. In practice, there are 95 such loans observed in our dataset, likely due to administrative errors. To adapt the [DeFusco et al. \(2020\)](#) methodology for the DTI>50 region, we set the counterfactual loan volume in the DTI>50 region equal to their pre-policy quantities:

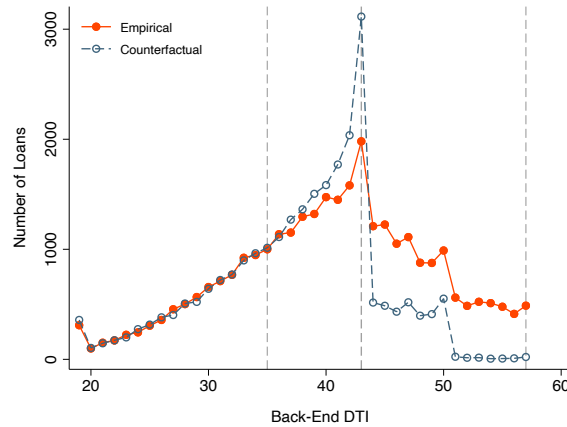
$$\hat{n}_{ld}^{post} = n_{ld}^{pre}, \text{ for } d > 50. \quad (12)$$

Figure [A.1](#) shows the bunching estimates using the adapted [DeFusco et al. \(2020\)](#) methodology. Inspecting Figure [A.1](#), we find that algorithms approved approximately as many loans in the DTI>50 region post-policy relative to the no-policy counterfactual in the 43–50 DTI region pre-policy, which suggests that algorithms were able to significantly expand credit supply in the DTI>50 region.

We then examine the delinquency risk of this purely algorithmically underwritten population, by comparing to DTI 43–50 low-FICO borrowers post-policy. Table [A.2](#) shows that the low-FICO borrowers in the DTI>50 region have even lower default risk than DTI 43–50 low-FICO borrowers post-policy, with 10% statistical significance in our fully saturated specification in column (3), suggesting that algorithms are able to significantly expand credit supply while controlling risk within the DTI>50 region, as consistent with our main results.

One explanation for the lower delinquency rates in the DTI>50 region post-policy, as compared to the DTI 43–50 region post-policy, is that algorithms may have been more restrictive with their credit supply in this region. This can be seen in the kink downwards in terms of a drop-off in the number of loans originated right after the DTI=50 cut-off in Figure [A.1](#). Interestingly, Figure [A.1](#) also shows that there is limited bunching at DTI=50 either pre-policy or post-policy, which suggests that unlike the DTI=43 point there may have been less intensive margin adjustment of DTI around DTI=50.

Figure A.1. Bunching analysis, including DTI>50 region with no human underwriting



Note: This figure plots the empirical and counterfactual distributions of FHA single-family, non-manufactured housing new purchase mortgages in our Ginnie Mae-Endorsements-HMDA sample 12 months after the policy, while including the $DTI > 50$ region. DTI is winsorized at the 1st and 99th percentiles and rounded up to the nearest integer. Dashed lines are drawn at DTI equals 43, above which the policy takes effect.

Table A.2. Delinquency Rates: DTI over 50

This table examines the changes in mortgage delinquency rates around the changes in underwriting regulations. The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017, excluding August 2016, the month of the policy implementation. The sample consists of borrowers with credit scores below 620, $DTI > 43$, and loan origination date after the regulation change in August 2016. $DTI > 50$ represents an indicator for borrowers with DTI greater than 50. Controls include log of loan amount and log of borrower household income. Standard errors are reported in parentheses and are double clustered by DTI (integer level) and origination month. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Delinquency Rates, within Post-Treatment Sample			
Sample	High DTI (> 43)		
Dep. Var.: <i>Delinquency</i>	(1)	(2)	(3)
<i>DTI > 50</i>	-0.00168 (0.00653)	-0.00609 (0.00647)	-0.0124* (0.00659)
Controls		Yes	Yes
Month FE	Yes	Yes	Yes
FICO FE	Yes	Yes	Yes
County FE			Yes
Lender FE			Yes
Observations	10779	10771	10075
R^2	0.002	0.006	0.149

C The Experian-Ginnie Mae match

We enrich the individual-year observations in the Experian data that had a new FHA mortgage origination with Ginnie Mae data, which contains more information about the details of the new FHA mortgage origination. We continue to use the FHA endorsements as an intermediate link, and use the Ginnie Mae-FHA endorsements data described in Section A as the starting point of the merge.

To conduct the merge, we use the following variables common to both datasets: the zip code of the property, the origination month, the original mortgage amount, and whether the borrower has co-applicants. In the Ginnie Mae-FHA endorsement data, these variables identify loans with 99.13% uniqueness. We drop the non-unique loans based on these variables in the Ginnie Mae-FHA endorsement data, and merge them with the Experian data.

The match rate, which we define as the fraction of loans within the Experian data that are merged to unique loans in the Ginnie Mae-FHA endorsement data, is given by year in Table A.3. A match rate below 62% is expected, due to the fact that the Ginnie Mae-endorsement data is only 62% of the original Ginnie Mae data as described in Section A. The match rates range between 41% to 53%, depending on year, which are not far from the maximum value of 62% on average. In the baseline analyses for Section 8.2, we weigh the individual-year observations with new FHA mortgages by the inverse of these year-specific match rates, effectively assuming that the matched sample is representative, in regression analyses that utilize merged variables. We also show results without matching and weighting in Table F.1.

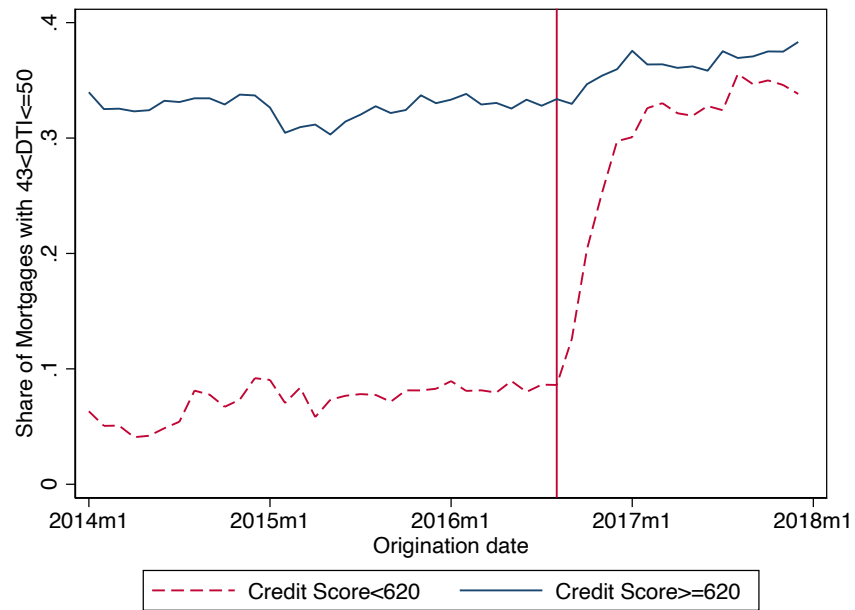
Table A.3. Match rate by year

Year	2014	2015	2016	2017	2018	2019
Match Rate	0.443	0.454	0.414	0.497	0.530	0.503

B Additional Results on Credit Quantity Effects

A Effect of the policy change in full sample

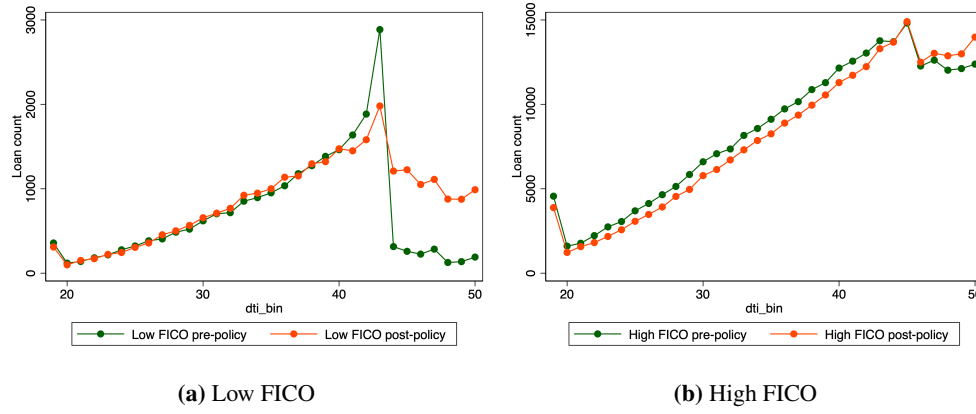
Figure B.1. Effect of the policy change, Ginnie Mae-HMDA sample



Note: This figure plots the share of FHA new purchase, single-family, non-manufactured housing mortgages with an DTI greater than 43 and less than or equal to 50 by their month of origination. The sample is the Ginnie Mae-HMDA sample from January 2015 to December 2017 with DTI less than or equal to 50. Data for borrowers with a credit score less than 620 and a credit score greater than or equal to 620 are separately plotted. The policy month of August 2016 is marked via a vertical red line.

B Raw data for Bunching Analysis

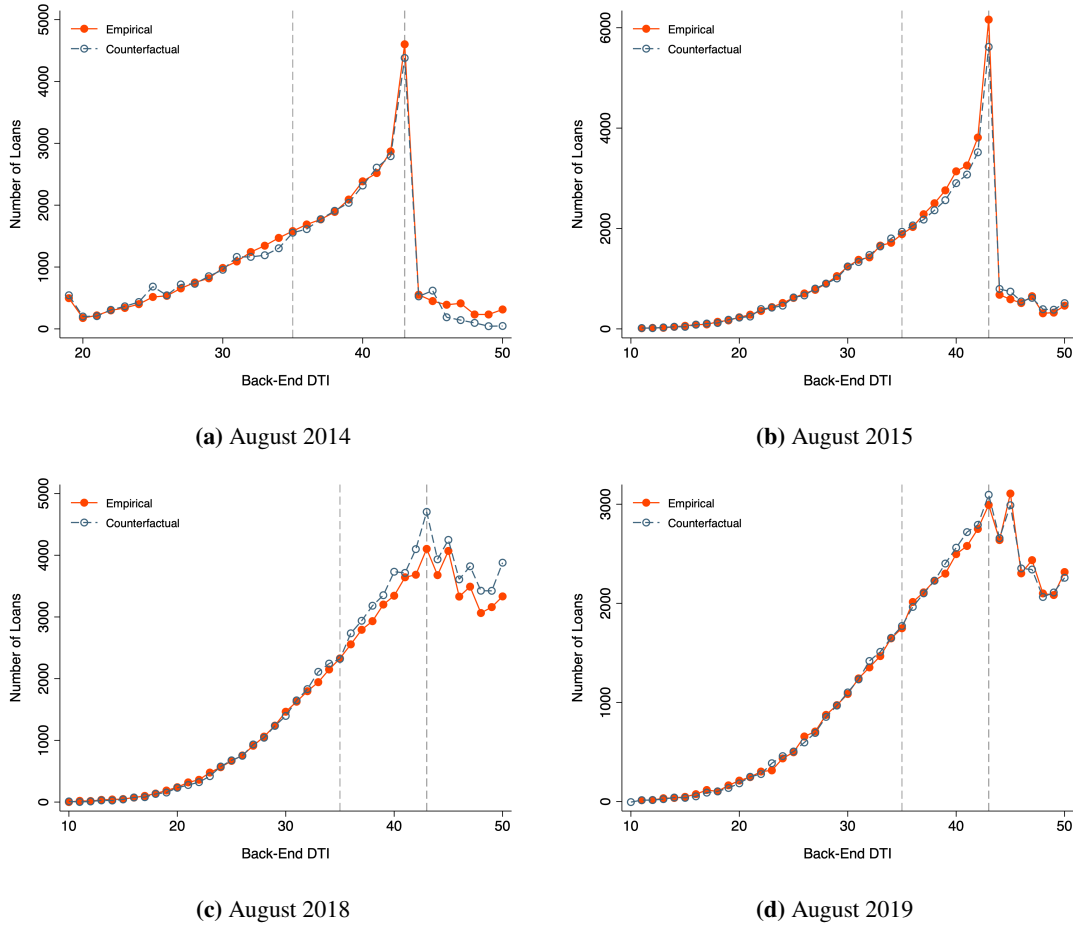
Figure B.2. Raw Data Input



Note: This figure plots empirical numbers of FHA single-family new purchase mortgages in our Ginnie Mae-Endorsements-HMDA sample pre- and-post policy, separately for low (below 620) and high (above 620) FICO borrowers. DTI is winsorized at the 1st and 99th percentiles and rounded up to the nearest integer.

C Placebo Bunching Analysis

Figure B.3. Placebo Bunching Analysis



Note: This figure plots empirical and counterfactual numbers of FHA single-family new purchase mortgages in our Ginnie Mae sample around various pseudo policy dates, based on the methodology described in Section 4.2. DTI is winsorized at the 1st and 99th percentiles and rounded up to the nearest integer. Dashed lines are drawn at DTI equals 43, above which the policy takes into affect, and at DTI equals 35, at or below which we assume is unaffected by the policy for our baseline bunching analysis. The sample consists of loans 12 months prior to and 12 months following the specified month, excluding the specified month.

C Additional Evidence for Delinquency Analysis

Table C.1. Delinquency Rates: Longer Time Horizon

This table examines the changes in mortgage delinquency rates around the changes in underwriting regulations. The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017, excluding August 2016, the month of the policy implementation. DTI is winsorized at the 1st and 99th percentiles and rounded up to the nearest integer. We report results for 3-year, 4-year, and 5-year delinquencies. *Treated* is an indicator that equals one if the borrower's credit score is below 620, and zero otherwise. *Post* indicates whether the loan is extended after the regulation change in August 2016. *High DTI* (*Low DTI*) represents a subsample of borrowers with DTI above 43 (less than or equal to 43). Borrowers with DTI below 35 are unaffected by the policy and are excluded from the sample. Controls include log of loan amount and log of borrower household income. Standard errors are reported in parentheses and are double clustered by DTI (integer level) and origination month. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Panel A. 3 Year Delinquency, Difference-in-difference Results						
Sample	High DTI ($43 < DTI \leq 50$)			Low DTI (≤ 43)		
Dep. Var.: <i>Delinquency</i> (3 year)	(1)	(2)	(3)	(4)	(5)	(6)
<i>Treated</i> $\times$ <i>Post</i>	-0.0112 (0.0140)	-0.0131 (0.0131)	-0.00620 (0.0143)	0.00581 (0.00684)	0.00601 (0.00708)	0.00808 (0.00755)
Controls		Yes	Yes		Yes	Yes
Month FE	Yes			Yes		
FICO FE	Yes			Yes		
FICO-DTI FE		Yes	Yes		Yes	Yes
Month-DTI FE		Yes	Yes		Yes	Yes
County FE			Yes			Yes
Lender FE			Yes			Yes
Observations	193092	192570	192226	203345	202698	202375
R^2	0.049	0.052	0.086	0.043	0.047	0.079

Panel B. 4 Year Delinquency, Difference-in-difference Results						
Sample	High DTI ($43 < DTI \leq 50$)			Low DTI (≤ 43)		
Dep. Var.: <i>Delinquency</i> (4 year)	(1)	(2)	(3)	(4)	(5)	(6)
<i>Treated</i> $\times$ <i>Post</i>	-0.00623 (0.0136)	-0.00769 (0.0134)	0.000996 (0.0159)	0.00428 (0.00663)	0.00403 (0.00659)	0.00679 (0.00749)
Controls		Yes	Yes		Yes	Yes
Month FE	Yes			Yes		
FICO FE	Yes			Yes		
FICO-DTI FE		Yes	Yes		Yes	Yes
Month-DTI FE		Yes	Yes		Yes	Yes
County FE			Yes			Yes
Lender FE			Yes			Yes
Observations	193092	192570	192226	203345	202698	202375
R^2	0.053	0.057	0.092	0.052	0.056	0.090

Panel C. 5 Year Delinquency, Difference-in-difference Results

Sample	High DTI ($43 < DTI \leq 50$)			Low DTI (≤ 43)		
Dep. Var.: <i>Delinquency (5 year)</i>	(1)	(2)	(3)	(4)	(5)	(6)
<i>Treated</i> $\times$ <i>Post</i>	-0.00816 (0.0137)	-0.00986 (0.0131)	-0.000102 (0.0157)	-0.00375 (0.00799)	-0.00414 (0.00786)	-0.00163 (0.00870)
Controls		Yes	Yes		Yes	Yes
Month FE	Yes			Yes		
FICO FE	Yes			Yes		
FICO-DTI FE		Yes	Yes		Yes	Yes
Month-DTI FE		Yes	Yes		Yes	Yes
County FE			Yes			Yes
Lender FE			Yes			Yes
Observations	193092	192570	192226	203345	202698	202375
R^2	0.045	0.049	0.088	0.047	0.052	0.089

Table C.2. Delinquency Rates: 30 and 60 Day Measures

This table examines the changes in mortgage delinquency rates around the changes in underwriting regulations. The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017, excluding August 2016, the month of the policy implementation. DTI is winsorized at the 1st and 99th percentiles and rounded up to the nearest integer. Panel A reports results from the DID analysis following Equation 7 for 2 year, 30-day delinquencies, and Panel B reports 2 year, 60-day delinquencies. *Treated* is an indicator that equals one if the borrower's credit score is below 620, and zero otherwise. *Post* indicates whether the loan is extended after the regulation change in August 2016. *High DTI* (*Low DTI*) represents a subsample of borrowers with DTI above 43 (less than or equal to 43). Borrowers with DTI below 35 are unaffected by the policy and are excluded from the sample. Controls include log of loan amount and log of borrower household income. Standard errors are reported in parentheses and are double clustered by DTI (integer level) and origination month. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Panel A. 30-Day Delinquency, Difference-in-difference Results

Sample	High DTI ($43 < DTI \leq 50$)			Low DTI (≤ 43)		
Dep. Var.: <i>Delinquency</i> (30 day)	(1)	(2)	(3)	(4)	(5)	(6)
<i>Treated</i> $\times$ <i>Post</i>	-0.000310 (0.0159)	0.00249 (0.0167)	0.00798 (0.0166)	0.00403 (0.00621)	0.00380 (0.00621)	0.00898 (0.00675)
Controls		Yes	Yes		Yes	Yes
Month FE	Yes			Yes		
FICO FE	Yes			Yes		
FICO-DTI FE		Yes	Yes		Yes	Yes
Month-DTI FE		Yes	Yes		Yes	Yes
County FE			Yes			Yes
Lender FE			Yes			Yes
Observations	193092	192570	192226	203345	202698	202375
R^2	0.065	0.069	0.105	0.071	0.076	0.111

Panel B. 60-Day Delinquency, Difference-in-difference Results

Sample	High DTI ($43 < DTI \leq 50$)			Low DTI (≤ 43)		
Dep. Var.: <i>Delinquency</i> (60 day)	(1)	(2)	(3)	(4)	(5)	(6)
<i>Treated</i> $\times$ <i>Post</i>	-0.0146 (0.0112)	-0.0147 (0.0105)	-0.0101 (0.0106)	0.00867 (0.00499)	0.00858 (0.00483)	0.0109 (0.00583)
Controls		Yes	Yes		Yes	Yes
Month FE	Yes			Yes		
FICO FE	Yes			Yes		
FICO-DTI FE		Yes	Yes		Yes	Yes
Month-DTI FE		Yes	Yes		Yes	Yes
County FE			Yes			Yes
Lender FE			Yes			Yes
Observations	193092	192570	192226	203345	202698	202375
R^2	0.042	0.045	0.082	0.043	0.047	0.082

Table C.3. Delinquency Rates: Excluding CRA tracts

This table examines the changes in mortgage delinquency rates around the changes in underwriting regulations. The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017, excluding August 2016, the month of the policy implementation. Low-income census tracts that qualify for the Community Reinvestment Act (CRA) assessment in the year of mortgage origination are excluded from this analysis. CRA data is obtained from [Chen et al. \(2025\)](#). DTI is winsorized at the 1st and 99th percentiles and rounded up to the nearest integer. *Treated* is an indicator that equals one if the borrower's credit score is below 620, and zero otherwise. *Post* indicates whether the loan is extended after the regulation change in August 2016. *High DTI* (*Low DTI*) represents a subsample of borrowers with DTI above 43 (less than or equal to 43). Borrowers with DTI below 35 are unaffected by the policy and are excluded from the sample. Controls include log of loan amount and log of borrower household income. Standard errors are reported in parentheses and are double clustered by DTI (integer level) and origination month. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Sample	High DTI ($43 < DTI \leq 50$)			Low DTI (≤ 43)		
Dep. Var.: <i>Delinquency</i> (90 day)	(1)	(2)	(3)	(4)	(5)	(6)
<i>Treated</i> $\times$ <i>Post</i>	-0.01000 (0.0103)	-0.0106 (0.0107)	-0.00872 (0.0106)	-0.00514 (0.00786)	-0.00511 (0.00816)	-0.00355 (0.00811)
Controls		Yes	Yes		Yes	Yes
Month FE	Yes			Yes		
FICO FE	Yes			Yes		
FICO-DTI FE		Yes	Yes			Yes
Month-DTI FE		Yes	Yes			Yes
County FE			Yes			Yes
Lender FE			Yes			Yes
Observations	156408	155958	155565	165756	165210	164858
R^2	0.030	0.033	0.072	0.029	0.033	0.070

D Additional Heterogeneity Analyses

A Propensity-Score Weighting of Race Subsamples by Income

This section provides more details regarding the covariance-balanced propensity score (CBPS) weighting approach used in Panel C of Table 5. The algorithm allows us to balance non-Black borrower’s adjusted income, i.e. application income relative to the area median (AMI), to match those of Black borrowers. Specifically, we follow the methodology of [Imai and Ratkovic \(2014\)](#) to generate weights based on following covariates X_i for each borrower i , separately pre-and-post policy:

$$X_i = \{\text{adjusted_income}_i, \text{adjusted_income_sq}_i, \text{Low FICO}_i, \\ \text{adjusted_income}_i \times \text{Low FICO}_i, \text{adjusted_income_sq}_i \times \text{Low FICO}_i\},$$

where adjusted_income_i and $\text{adjusted_income_sq}_i$ are continuous variables representing the adjusted income of the borrower and its squared term. Low FICO_i is an indicator for whether the borrower has $\text{FICO} < 620$. From our regression sample, 0.22% of Black borrowers and 0.30% of white borrowers are not assigned a weight and are excluded from the sample. Table D.1 shows that the distributions of the adjusted income of Black and non-Black borrowers are well-balanced after weighting, both across low- and high-FICO groups and pre- and post-policy.

Table D.1. Post-Weighting Summary Statistics on adjusted_income_i by FICO Group and Race

Panel A: Pre-policy								
	Race	Mean	SD	P10	P25	Median	P75	P90
Low FICO	Black	1.084	0.604	0.567	0.726	0.981	1.321	1.700
	Non-Black	1.084	0.604	0.561	0.730	0.987	1.311	1.685
High FICO	Black	0.953	0.478	0.470	0.617	0.846	1.173	1.563
	Non-Black	0.953	0.478	0.468	0.611	0.851	1.186	1.562

Panel B: Post-policy								
	Race	Mean	SD	P10	P25	Median	P75	P90
Low FICO	Black	1.092	0.486	0.583	0.747	0.998	1.333	1.709
	Non-Black	1.092	0.486	0.579	0.751	1.009	1.326	1.701
High FICO	Black	0.971	0.486	0.484	0.632	0.866	1.193	1.592
	Non-Black	0.971	0.486	0.482	0.630	0.873	1.205	1.577

B Nonparametric Weighting of Race Subsamples by Income

We provide results on an alternative matching approach. Instead of re-weighting each individual loan by their propensity scores, we compute weights from a non-parametric method by FICO-income grids. Specifically, we group adjusted household income into buckets of 0.1, and for each FICO score-income bucket, we compute weights w as the relative likelihood that the bucket contains non-Black borrowers relative to black borrowers ($w = \frac{\#NonBlack}{Black}$). This weight is assigned to non-Black borrowers in that FICO-income grid, and can be viewed as a population weights that “correct” the density distribution of non-Black borrowers in that grid, so their overall income distribution becomes identical to that of Black borrowers. Loans to Black borrowers are assigned a weight of 1.

Table D.2. Heterogeneity by Race, Non-parametric Weighting by Income

This table examines the heterogeneity effects of the FHA underwriting policy across racial and income groups. The key variables are defined in the same way as in Table 5. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Panel D: Race Partition After Non-Parametric Weighting by Income			
Subsample:	Black	Non-Black	Difference (Black - Non-Black)
Outcome: <i>Volume Change</i>			
<i>ΔLoans Originated</i>	-0.010 (0.038)	0.071*** (0.012)	-0.081** (0.038)
<i>g(Volume) (High DTI)</i>	0.859*** (0.151)	1.218*** (0.071)	-0.359** (0.162)
Outcome: <i>Delinquency Change (Coeff on Treated × Post)</i>			
High DTI:	0.026 (0.029)	-0.014 (0.012)	0.041 (0.027)
Low DTI:	-0.000 (0.012)	0.002 (0.006)	-0.002 (0.013)

C Additional heterogeneity analysis by race and ethnicity

Table D.3. Change in FICO by Race and Income

This table examines the changes in observed FICO scores around the changes in underwriting regulations. The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017, excluding August 2016, the month of the policy implementation. The sample consists of borrowers with DTI between 43 and 50. Controls consist of the dummy variable indicating whether the loan is originated by the treated group of borrowers with FICO<620. Standard errors are reported in parentheses and are double clustered by DTI (integer level) and origination month. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Panel A: Race			
Sample:	High DTI ($43 < DTI \leq 50$)		
Dep. Var.: <i>FICO</i>	(1)	(2)	(3)
Subsample:	Black	Non-Black	Difference (Black - Non-Black)
<i>Treated × Post</i>	1.941 (1.829)	5.686*** (1.220)	-3.744* (1.814)
Controls	Yes	Yes	–
County FE	Yes	Yes	–
Lender FE	Yes	Yes	–
Month FE	Yes	Yes	–
Observations	24,615	168,212	–
R^2	0.285	0.195	–

Panel B: Income			
Sample:	High DTI ($43 < DTI \leq 50$)		
Dep. Var.: <i>FICO</i>	(1)	(2)	(3)
Subsample:	Low Income	High Income	Difference (Low - High Income)
<i>Treated × Post</i>	5.772*** (1.596)	4.619*** (1.125)	1.259 (1.227)
Controls	Yes	Yes	–
County FE	Yes	Yes	–
Lender FE	Yes	Yes	–
Month FE	Yes	Yes	–
Observations	109,774	82,908	–
R^2	0.182	0.252	–

Table D.4. Heterogeneity by Ethnicity

This table examines the changes in loan origination volume and delinquency rates around the changes in underwriting regulations for Hispanic and non-Hispanic borrowers. $\Delta\text{Loans Originated}$ refers to the increase in the total number of new purchase loans extended to low FICO borrowers as a fraction of the number of new purchase loans extended to low FICO borrowers in the absence of the policy. $g(\text{Volume}) (\text{High DTI})$ refers to the increase in the total number of new purchase loans extended to borrowers with FICO below 620 and DTI within 43–50 after the policy change as a fraction of the counterfactual number of new purchase loans for those borrowers in the absence of the policy. The methodology is described in Section 4.2. *Difference* represents the difference in coefficient estimates between each subsample partition and the statistical significance of the gap. The methodology is described in Section 4.2. The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017, excluding August 2016, the month of the policy implementation. DTI is winsorized at the 1st and 99th percentiles and rounded up to the nearest integer. Standard errors are reported in parentheses and are from 1,000 bootstrap replications clustered by origination month for the bunching analyses, and two-way clustered by origination month and DTI for regression analyses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

	Hispanic	Non-Hispanic	Difference (Hispanic - Non-Hispanic)
<i>Volume Change</i>			
$\Delta\text{Loans Originated}$	0.077* (0.041)	0.063*** (0.013)	0.014 (0.042)
$g(\text{Volume})(\text{HighDTI})$	1.224*** (0.202)	1.207*** (0.080)	0.017 (0.218)
<i>Delinquency Change (Coeff on Treated X Post)</i>			
High DTI:	-0.046 (0.026)	-0.000 (0.012)	-0.045* (0.023)
Low DTI:	-0.012 (0.015)	0.003 (0.007)	-0.016 (0.016)

D Propensity score weighting of lender type and bank size by indirect seller status

In Panel B of Table 7, we use the covariance-balanced propensity score (CBPS) weighting approach to balance indirect sellers' lender type (i.e., bank vs non-bank) and bank size (i.e., asset size) to match those of direct sellers. We use the methodology of Imai and Ratkovic (2014) to generate weights based on following covariates X_i for each borrower i , separately pre-and-post policy:

$$X_i = \{\text{bank}_i, \text{bank}_i \times \log_assets_i, \text{bank}_i \times \log_assets_sq_i, \text{Low FICO}_i, \text{bank}_i \times \text{Low FICO}_i, \text{bank}_i \times \log_assets_i \times \text{Low FICO}_i, \text{bank}_i \times \log_assets_sq_i \times \text{Low FICO}_i\},$$

where bank_i is an indicator variable for whether the lender is a bank originator, $\log_assets_i$, $\log_assets_sq_i$ are continuous variables representing the bank's 2015 log asset size and square of the bank's 2015 asset size, respectively, and Low FICO_i is a dummy variable indicating whether the borrower has $\text{FICO} < 620$. Non-banks are assigned a 2015 asset size of 0. Table D.5 shows the results of the weighting in terms of the lender type in Panel A and the mean and standard deviation of asset size among bank lenders in Panel B.

Table D.5. Post-Weighting Summary Statistics on lender type and asset size by FICO Group and Indirect Seller Status

Panel A. Lender type (bank_i)					
FICO Group	Seller Type	Pre-Policy Mean		Post-Policy Mean	
Low FICO	Indirect Sellers	0.175		0.158	
	Other Originators	0.175		0.158	
High FICO	Indirect Sellers	0.359		0.325	
	Other Originators	0.359		0.325	

Panel B. Asset Size ($\log_assets_i$), Among Bank Lenders					
FICO Group	Indirect Sellers	Pre-Policy		Post-Policy	
		Mean	SD	Mean	SD
Low FICO	Indirect Sellers	13.806	1.302	13.816	1.292
	Other Originators	13.806	1.302	13.816	1.292
High FICO	Indirect Sellers	14.164	1.465	14.054	1.441
	Other Originators	14.164	1.465	14.054	1.441

E Propensity score weighting of borrower heterogeneity by borrower race and income

In this section we use the covariance-balanced propensity score (CBPS) weighting approach to balance the race (i.e., Black and non-black) and adjusted income of: (1) first-time home buyers and non-first time home buyers, (2) sole-borrowers and borrowers with co-applicants, and (3) borrowers in high self-employment zip codes and borrowers in low self-employment zip codes. We use the methodology of [Imai and Ratkovic \(2014\)](#) to generate weights based on following covariates X_i for each borrower i , separately pre-and-post policy:

$$X_i = \text{race}_i, \text{race}_i \times \text{adjusted_income}_i, \text{race}_i \times \text{adjusted_income_sq}_i, \text{Low FICO}_i, \\ \text{race}_i \times \text{Low FICO}_i, \text{race}_i \times \text{adjusted_income}_i \times \text{Low FICO}_i, \text{race}_i \times \text{adjusted_income_sq}_i \times \text{Low FICO}_i,$$

where race_i is an indicator variable for whether the borrower is Black, adjusted_income_i , $\text{adjusted_income_sq}_i$ are continuous variables representing the application household income divided by the MSA median its square, and Low FICO_i is a dummy variable indicating whether the borrower has $\text{FICO} < 620$. Tables [D.6](#) to [D.8](#) show the results of the weighting on race and income.

Table D.6. Post-Weighting Summary Statistics on Race and Income by First Time Home Buyer Status

Panel A. Race (black_i)					
FICO Group	First-time Home Buyers	Pre-Policy Mean		Post-Policy Mean	
Low FICO	No	0.171		0.119	
	Yes	0.171		0.119	
High FICO	No	0.183		0.132	
	Yes	0.183		0.132	

Panel B. Adjusted Income					
FICO Group	First-time Home Buyers	Pre-Policy		Post-Policy	
		Mean	SD	Mean	SD
Low FICO	No	1.109	0.490	0.969	0.479
	Yes	1.109	0.490	0.969	0.479
High FICO	No	1.102	0.472	0.984	0.477
	Yes	1.102	0.472	0.984	0.477

Table D.7. Post-Weighting Summary Statistics on Race and Income by Co-Borrower Status

Panel A. Race (black_i)					
FICO Group	Has Co-Borrower	Pre-Policy Mean		Post-Policy Mean	
Low FICO	Yes	0.179		0.132	
	No	0.179		0.132	
High FICO	Yes	0.189		0.146	
	No	0.189		0.146	

Panel B. Adjusted Income					
FICO Group	Has Co-Borrower	Pre-Policy		Post-Policy	
		Mean	SD	Mean	SD
Low FICO	Yes	1.019	0.465	0.877	0.429
	No	1.019	0.465	0.877	0.429
High FICO	Yes	1.014	0.450	0.889	0.426
	No	1.014	0.450	0.889	0.426

Table D.8. Post-Weighting Summary Statistics on Race and Income by Self-Employment Zip Code

Panel A. Race (black_i)					
FICO Group	Self-Employment in Zip Code	Pre-Policy Mean		Post-Policy Mean	
Low FICO	Low	0.154		0.105	
	High	0.154		0.105	
High FICO	Low	0.162		0.115	
	High	0.162		0.115	

Panel B. Adjusted Income					
FICO Group	Self-Employment in Zip Code	Pre-Policy		Post-Policy	
		Mean	SD	Mean	SD
Low FICO	Low	1.216	0.541	1.083	0.531
	High	1.216	0.541	1.083	0.531
High FICO	Low	1.198	0.519	1.094	0.524
	High	1.198	0.519	1.094	0.524

F Bunching subsample graphs

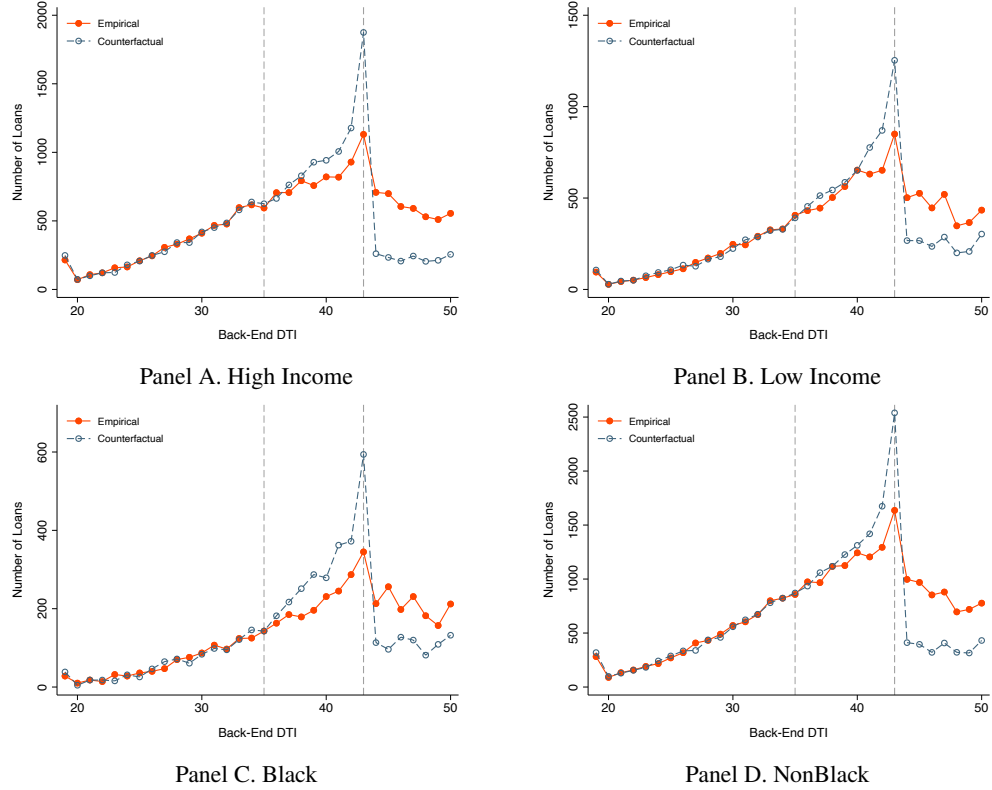


Figure D.1. This figure presents heterogeneous changes in credit quantity and DTI distribution around the FHA policy adoption for subsamples of borrower income and race. Panels A and B provide the DTI distributions for borrowers with above- and below-median income, respectively. Panels C and D provide the DTI distributions for Black and non-Black borrowers, respectively. These graphs correspond to the subsample analyses in Table 5

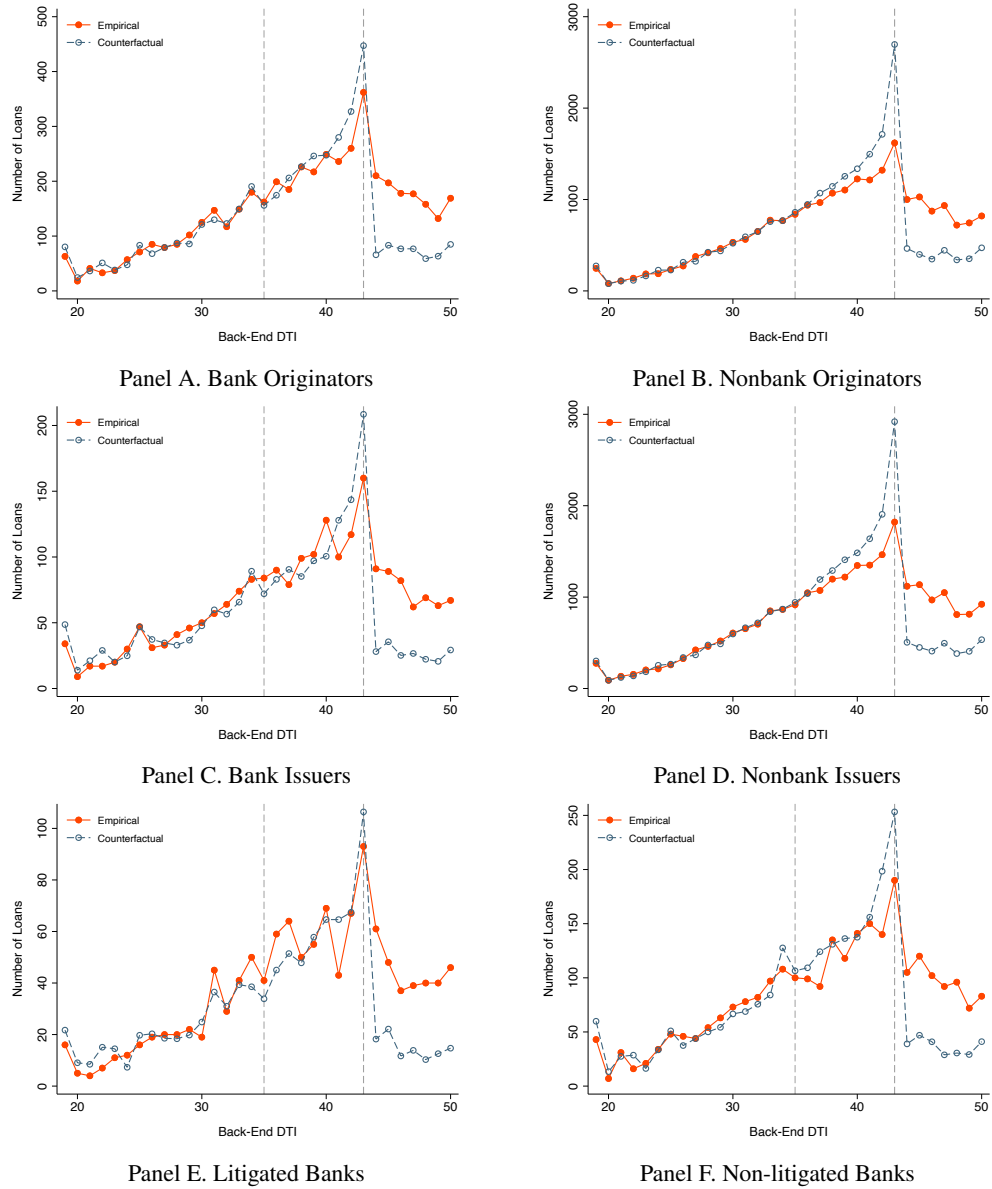
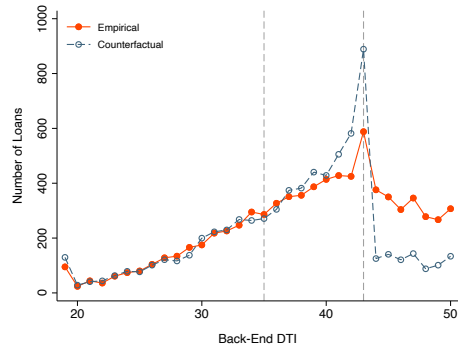
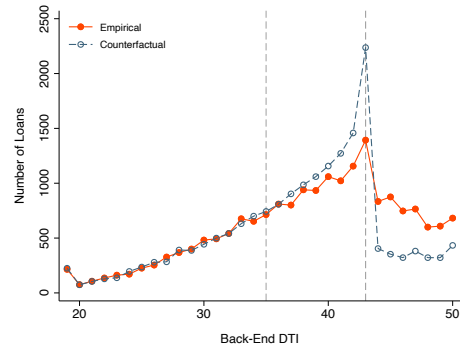


Figure D.2. This figure presents the heterogeneous changes in credit quantity and DTI distribution around the FHA policy adoption for subsamples based on lenders' concerns regarding regulatory putback or unobservable human underwriting quality. Panels A and B provide the DTI distributions for bank and nonbank originators, respectively. Panels C and D provide the DTI distributions for bank and nonbank issuers, respectively. Panels E and F provide the DTI distributions for lenders that have and have not faced litigation regarding fraudulent FHA practices, respectively. These graphs correspond to the subsample analyses in Table 6.



Panel A. Indirect Sellers



Panel B. Other Originators

Figure D.3. This figure presents heterogeneous changes in credit quantity and DTI distribution around the FHA policy adoption for subsamples based on lenders' concerns regarding unobservable human underwriting quality. Panels A and B compare indirect sellers with other originators, respectively. These graphs correspond to the subsample analyses in Table 7.

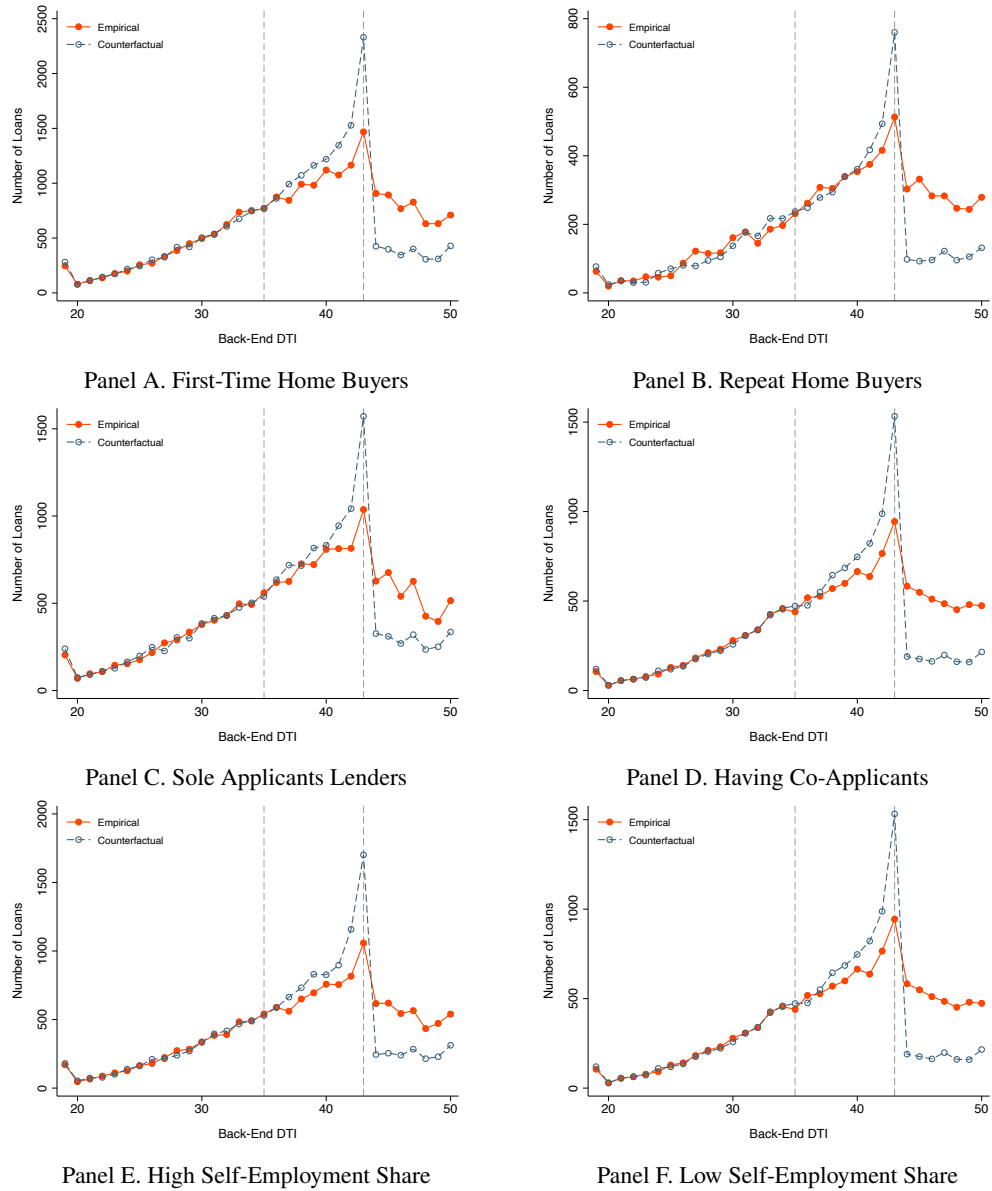


Figure D.4. This figure depicts the heterogeneous changes in credit quantity and DTI distribution around the FHA policy adoption across borrower characteristics. Panels A and B provide the DTI distributions for first-time and repeat home buyers, respectively. Panels C and D report the results for sole applicants and borrowers with co-applicants, respectively. Panels E and F report the results for borrowers in zip codes with above-median and below-median shares of self-employment income, respectively. These graphs correspond to the subsample analyses in Table 8.

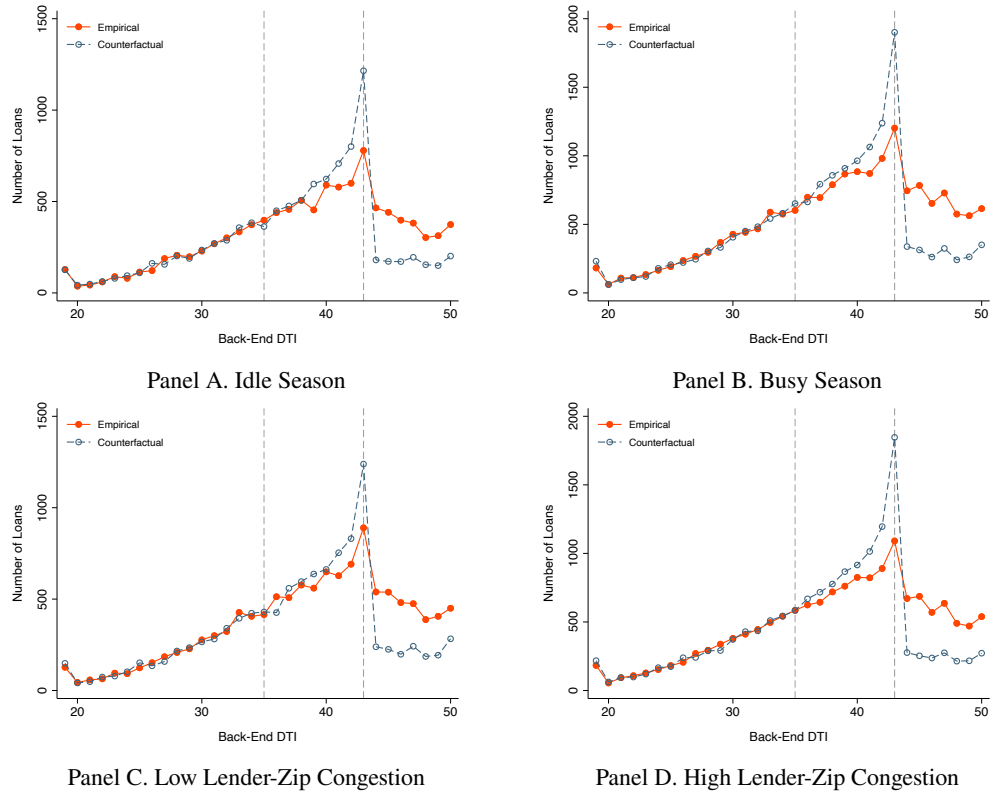


Figure D.5. This figure presents heterogeneous changes in credit quantity and DTI distribution around the FHA policy adoption for subsamples related to lender capacity constraints. Panels A and B provide the DTI distributions for idle and busy seasons, respectively. Panels C and D provide the DTI distributions for low and high congestion in a lender-zip code, respectively. These graphs correspond to the subsample analyses in Table E.1.

E Additional Evidence on Economic Mechanisms

Table E.1. Mechanism: Lender Capacity Constraints

This table examines the heterogeneity effects of the FHA underwriting policy across areas and periods with high and low loan application growth. $\Delta\text{Loans Originated}$ refers to the increase in the total number of new purchase loans extended to low FICO borrowers as a fraction of the counterfactual number of new purchase loans extended to low FICO borrowers in the absence of the policy. $g(\text{Volume})$ (*High DTI*) refers to the increase in the total number of new purchase loans extended to borrowers with FICO below 620 and DTI within 43–50 after the policy change as a fraction of the counterfactual number of new purchase loans for those borrowers in the absence of the policy. The methodology is described in Section 4.2. *Difference* represents the difference in coefficient estimates between each subsample partition and the statistical significance of the gap. Panel A reports results where we partition the sample based on months with low and high application volumes. March through August are classified as the “High Season”, while September through February are “Low Season.” Panel B reports results where we partition the sample based on lender-zips with above-median (“High Congestion”) and below-median growth rates of application volume (“Low Congestion”). The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017, excluding August 2016, the month of the policy implementation. DTI is restricted to between 35 to 50. Standard errors are reported in parentheses and are from 1,000 bootstrap replications clustered by origination month. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Panel A: Seasonality			
Subsample:	Idle Season	Busy Season	Difference (Idle – Busy)
Outcome: <i>Volume Change</i>			
$\Delta\text{Loans Originated}$	0.053** (0.025)	0.071*** (0.015)	-0.018 (0.029)
$g(\text{Volume})$ (<i>High DTI</i>)	1.186*** (0.138)	1.229*** (0.095)	-0.043 (0.167)
Panel B: Lender-Zip Congestion			
Subsampler:	Low Congestion	High Congestion	Difference (Low – High)
Outcome: <i>Volume Change</i>			
$\Delta\text{Loans Originated}$	0.093*** (0.020)	0.054** (0.026)	0.039 (0.029)
$g(\text{Volume})$ (<i>High DTI</i>)	1.091*** (0.105)	1.224*** (0.136)	-0.133 (0.148)

Table E.2. Robustness Tests for Lender Capacity Constraints

This table reports the results from alternative specifications related to measuring lenders' capacity constraints. $g(\text{Volume})$ (*High DTI*) refers to the increase in the total number of new purchase loans extended to borrowers with FICO below 620 and DTI within 43–50 after the policy change as a fraction of the counterfactual number of new purchase loans for those borrowers in the absence of the policy. The methodology is described in Section 4.2. *Difference* represents the difference in coefficient estimates between each subsample partition and the statistical significance of the gap. Panel A reports results where we partition the sample based on lender-tract with above-median and below-median growth rates of application volume; and Panel B reports results where we measure lender-zip congestion using loan application volume excluding FHA applications. The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017. DTI is restricted to between 35 to 50. Standard errors are reported in parentheses and are from 1,000 bootstrap replications clustered by origination month. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Panel A: Lender-Tract Congestion			
	Low Lender-Tract Congestion	High Lender-Tract Congestion	Difference
<i>Volume Change</i>			
$\Delta\text{Loans Originated}$	0.102*** (0.024)	0.062*** (0.023)	0.040 (0.025)
$g(\text{Volume})$ (High DTI)	1.180*** (0.126)	1.210*** (0.117)	-0.030 (0.148)
Panel B: Lender-ZIP Congestion, Excluding FHA Applications			
	Low Lender-Zip Congestion	High Lender-Zip Congestion	Difference
<i>Volume Change</i>			
$\Delta\text{Loans Originated}$	0.095*** (0.029)	0.070*** (0.025)	0.025 (0.034)
$g(\text{Volume})$ (High DTI)	1.196*** (0.123)	1.275*** (0.105)	-0.079 (0.130)

Table E.3. Mechanism: Lender Entry

This table reports the share of low FICO, high DTI loans post-policy that were made by existing versus new lenders in this market. The sample is our Ginnie Mae-HMDA sample of FHA single-family, non-manufactured housing, home purchase mortgages issued during the period of August 2015 through August 2017. DTI is restricted to between 35 to 50. Standard errors are reported in parentheses and are double clustered by DTI (integer level) and origination month. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

	(1) Existing Lenders	(2) New Lenders
<i>Share of low FICO, high DTI loans</i>	0.805*** (0.004)	0.195*** (0.004)
Number of Lenders	224	394

F Additional Evidence on Neighborhood Quality

Table F.1. Mortgage Access and Neighborhood Choices: Unweighted

This table uses 2SLS specifications to examine the effect of mortgage access on the changes in individuals' local school district ratings. The sample includes individuals in the 1% national representative sample of credit bureau annual records from 2014 to 2019 (excluding 2016), and is merged with the school rating data based on the location of individuals. The unit of observation is an individual-year. Panel A reports first-stage estimates where the dependent variable is an indicator *New Purchase FHA* that equals one if an individual has obtained a new FHA mortgage purchase in a given year. Panel B reports second-stage estimates of the new FHA mortgage purchase on changes in upward mobility, computed as the year-on-year difference. *Treated (2015)* is an indicator that equals one if the borrower's credit score is below 620 in 2015, and zero otherwise. *Post* indicates whether the loan is extended after the regulation change in 2016. Individual characteristics include indicators for gender, marital status, and *Treated (2015)*. Age groups are: 18–35, 35–45, and above 45. A placebo test of this analysis for renters is shown in Appendix Table F.3. Standard errors are reported in parentheses and are clustered by county. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Panel A. First Stage, Obtaining FHA Mortgage				
Dep. Var.: <i>New Purchase FHA</i>	(1)	(2)	(3)	
<i>Treated (2015) × Post</i>	0.0019*** (0.0001)	0.0019*** (0.0001)	0.0019*** (0.0001)	
Individual Char	Yes	Yes	Yes	
Year FE	Yes			
Credit Score FE	Yes	Yes	Yes	
Zip Code FE	Yes			
Age Group FE		Yes		
Zip-Year FE		Yes	Yes	
Age Group-Year FE			Yes	
Married-Year FE			Yes	
Gender-Year FE			Yes	
Observations	10,698,445	10,698,445	10,698,445	
R ²	0.01	0.01	0.03	
F-statistic	380.40	341.31	297.58	

Panel B. Moving to Opportunity				
Dep. Var:	<i>d(Up Mob, P25)</i>		<i>d(Up Mob, P50)</i>	
	(1)	(2)	(3)	(4)
<i>New Purchase FHA</i>	0.151*** (0.024)	0.129*** (0.024)	0.151*** (0.023)	0.131*** (0.023)
Individual Char	Yes	Yes	Yes	Yes
Credit Score FE	Yes	Yes	Yes	Yes
Age Group FE	Yes		Yes	
Zip-Year FE	Yes	Yes	Yes	Yes
Age Group-Year FE		Yes		Yes
Married-Year FE		Yes		Yes
Gender-Year FE		Yes		Yes
Observations	10,670,325	10,670,325	10,670,325	10,670,325

Table F.2. Mortgage Access and Neighborhood Choices: Other Mobility Measures

This table reports the second-stage estimates of the 2SLS approach, which examines the effect of mortgage access on “moving to opportunity.” The sample includes individuals in the 1% national representative sample of credit bureau annual records from 2014 to 2019 (excluding 2016), and is merged with the upward mobility and neighborhood characteristics measures obtained from Chetty et al. (2025) based on the location of individuals. We exclude individuals that have obtained a mortgage with DTI above 50 in the past. The unit of observation is an individual-year. The dependent variables represent year-on-year changes in upward mobility of an individual’s census tract of residence. *School Rating* measures the public school district rating of the zip code. *Household Income* measures average household income. *Poverty Share* measures the share of individuals in the tract below the federal poverty line. *Frac College* measures the number of people aged 25 or older who have a bachelor’s degree, master’s degree, professional school degree, or doctorate degree, divided by the total number of people aged 25 or older in a tract. *Single Parent Share* measures the number of households with female heads (and no husband present) or male heads (and no wife present) with own children under 18 years old present divided by the total number of households with own children present. *Treated (2015)* is an indicator that equals one if the borrower’s Vantage Score is below 620 in 2015, and zero otherwise. *Post* indicates whether the loan is extended after the regulation change in 2016. Individual characteristics include indicators for gender, marital status, and *Treated (2015)*. Age groups are: 18–35, 35–45, and above 45. Standard errors, clustered by county, are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Dep. Var:	<i>d(School Rating)</i>		<i>d(Household Income)</i>		<i>d(Poverty Share)</i>		<i>d(Frac College)</i>		<i>d(Single Parent Share)</i>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>New Purchase FHA</i>	1.2693** (0.5440)	1.1906** (0.5047)	36,970.908*** (13,931)	13,633.639 (12,875)	-0.188*** (0.060)	-0.109** (0.054)	0.599*** (0.081)	0.517*** (0.074)	-0.282*** (0.101)	-0.162* (0.091)
Individual Char	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Credit Score FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Age Group FE	Yes		Yes		Yes		Yes		Yes	
Year-Zip FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-Gender FE		Yes		Yes		Yes		Yes		Yes
Year-Age Group FE		Yes		Yes		Yes		Yes		Yes
Year-Marriage Status FE		Yes		Yes		Yes		Yes		Yes
Observations	10,661,657	10,661,657	10,662,328	10,662,328	10,662,706	10,662,706	10,640,338	10,640,338		

Table F.3. Mortgage Access and the Quality of Neighborhoods: Placebo Test Using Renters

The unit of observation is an individual-year. The sample is limited to individuals in the Experian dataset with a consistent dwelling status of "A" (indicating residence in apartments, condominiums, or other multi-family units) throughout the sample period. This selection criterion helps identify likely renters, following the approach of [Butler et al. \(2019\)](#). The outcome variables, specified at the top of each column, are detailed in the notes to Tables 10 and F.2. *Treated (2015)* is an indicator that equals one if the borrower's credit score is below 620 in 2015, and zero otherwise. *Post* indicates whether the loan is extended after the regulation change in 2016. Individual characteristics include indicators for gender, marital status, and *Treated (2015)*. The fixed effects specification is the same as the specification of the even-numbered columns in Table 10. Standard errors are reported in parentheses and are clustered by county. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Dep. Var:	(1) <i>d(School Rating)</i>	(2) <i>d(Up Mob, P25)</i>	(3) <i>d(Up Mob, P50)</i>	(4) <i>d(Household Income)</i>	(5) <i>d(Poverty Share)</i>	(6) <i>d(Frac College)</i>	(7) <i>d(Single Parent Share)</i>
<i>Treated (2015) × Post</i>	-0.004 (0.003)	-0.000 (0.000)	0.000 (0.000)	61.519 (103.606)	-0.000 (0.000)	-0.000 (0.000)	-0.001 (0.001)
Individual Char	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Credit Score FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-Zip FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-Gender FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-Age FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-Marriage Status FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	480,174	479,304	479,304	479,827	479,925	479,946	476,911
R ²	0.126	0.095	0.091	0.097	0.100	0.093	0.094